Design for Circularity in Steel Door Manufacturing

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Production and Logistics

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ABSTRACT

Level: Master thesis in Product and Process Development, advanced level, 30 credits.

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Title: Design for Circularity in Steel Door Manufacturing

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Aim: The aim of this study is to map the current design requirements and practices that influence steel safety door design and to develop recommendations for improving circularity in future steel door systems.

Research questions: RQ1: What current design requirements and practices influence steel door design, and how do they support or limit circularity?

RQ2: How can circular strategies be translated into design recommendations for future steel safety doors, including end-of-life considerations, given the identified requirements and constraints?

Methods: This study is based on qualitative research using an abductive approach. Empirical data was collected through meetings, a production walkthrough, semi-structured interviews with actors connected to the steel door value chain, and benchmarking of circularity practices in the steel door sector. The analysis followed a Gioia-inspired methodology to identify first-order concepts, second-order themes and aggregate dimensions. The empirical findings were analyzed in relation to circular economy, the 10R hierarchy, Design for X, modularity, traceability and product information.

Conclusion: For the first research question, the study identified that steel safety door design is strongly influenced by certification, fire safety requirements, durability demands, production practices, material selection and formal product compliance. These requirements support circularity through long service life, robustness, maintenance and selected component replacement, but they also limit higher-value circular strategies such as reuse, refurbishment and remanufacturing when certified performance cannot be verified. For the second research question, the study presents design recommendations focused on product information, traceability, maintenance support, replaceable wear components, controlled disassembly and clearer boundaries between safety-critical and replaceable components. The study concludes that circularity in certified steel safety doors should be developed step by step, starting with improved documentation and traceability before more advanced circular strategies such as reuse, refurbishment and remanufacturing are implemented.

Keywords: Circular economy, steel safety doors, design for circularity, Design for X, traceability, 10R hierarchy, product information, certification

SAMMANFATTNING

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Syfte: Syftet med denna studie är att kartlägga de nuvarande designkrav och arbetssätt som påverkar utformningen av stålsäkerhetsdörrar samt att utveckla rekommendationer för att förbättra cirkularitet i framtida ståldörrssystem.

Forskningsfrågor: FF1: Vilka nuvarande designkrav och arbetssätt påverkar utformningen av ståldörrar, och hur stödjer eller begränsar de cirkularitet?

FF2: Hur kan cirkulära strategier översättas till designrekommendationer för framtida stålsäkerhetsdörrar, inklusive hänsyn till slutet av livscykeln, utifrån de identifierade kraven och begränsningarna?

Metod: Denna studie bygger på kvalitativ forskning med en abduktiv ansats. Empiriska data samlades in genom möten, en produktionsgenomgång, semistrukturerade intervjuer med aktörer kopplade till ståldörrens värdekedja samt benchmarking av cirkularitetsrelaterade arbetssätt inom ståldörrssektorn. Analysen följde en Gioia-inspirerad metod för att identifiera första ordningens begrepp, andra ordningens teman och aggregerade dimensioner. De empiriska resultaten analyserades i relation till cirkulär ekonomi, 10R hierarkin, Design for X, modularitet, spårbarhet och produktinformation.

Slutsats: För den första forskningsfrågan identifierade studien att utformningen av stålsäkerhetsdörrar påverkas starkt av certifiering, brandsäkerhetskrav, hållbarhetskrav, produktionsmetoder, materialval och formell produktöverensstämmelse. Dessa krav stödjer cirkularitet genom lång livslängd, robust konstruktion, underhåll och utbyte av vissa komponenter, men de begränsar även mer värdebevarande cirkulära strategier såsom återbruk, renovering och återtillverkning när certifierad prestanda inte kan verifieras. För den andra forskningsfrågan presenterar studien designrekommendationer med fokus på produktinformation, spårbarhet, stöd för underhåll, utbytbara slitagedelar, kontrollerad demontering och tydligare gränser mellan säkerhetskritiska och utbytbara komponenter. Studien drar slutsatsen att cirkularitet i certifierade stålsäkerhetsdörrar bör utvecklas stegvis, med förbättrad dokumentation och spårbarhet som första steg innan mer avancerade cirkulära strategier såsom återbruk, renovering och återtillverkning implementeras.

Nyckelord: Cirkulär ekonomi, stålsäkerhetsdörrar, design för cirkularitet, Design for X, spårbarhet, 10R hierarkin, produktinformation, certifiering

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Contents

1. INTRODUCTION 1

1.1 Background 1
1.2 Problem Statement 3
1.3 Aim and Research questions 4
1.4 Project limitations 4

2. THEORETICAL FRAMEWORK 5

2.1 Design for X 5
2.1.1 Design for Disassembly 6
2.1.2 Design for Maintenance 6
2.1.3 Design for Remanufacturing 7
2.2 Circular Design Strategies for Value Retention 8
2.2.1 Resource loops 9
2.3 10R Hierarchy 10
2.4 Modularity in industrial products 12
2.5 Traceability 13

3. RESEARCH METHOD 14

3.1 Methodological approach 14
3.2 Research design 14
3.3 Case description 15
3.4 Data collection 16
3.4.1 Primary data collection 16
3.4.2 Secondary data collection 18
3.4.2.1 Benchmarking 19
3.5 Data analysis 19
3.6 Research quality 20

4. RESULTS 21

4.1 Benchmarking Results 21
4.2 Regulatory and production constraints 23
4.3 Material and maintenance conditions 24
4.4 System-level conditions 26
4.5 Future circular design requirements 28
4.6 Stakeholder Mapping 29
4.7 Summary framework of circularity possibilities and design recommendations 30

5. ANALYSIS AND DISCUSSION 32

5.1 Design requirements, Industry Practices, and Circularity in Steel Doors 32
5.1.1 Certification and performance requirements 32
5.1.2 Production practices and product architecture 33
5.1.3 Material and maintenance conditions 34
5.1.4 System-level conditions and stakeholder coordination 35
5.2 Design Recommendations for Circular steel safety doors 36
5.2.1 Verified performance as a starting point for circular design 36
5.2.2 Product architecture and intervention boundaries 37
5.2.3 Maintenance, service, and life extension during use 38
5.2.4 Product information, traceability, and material transparency 38
5.2.5 Disassembly and end-of-life handling 39
5.3 Practical implications 40
5.3.1 Implications for CH Industry 40

6. CONCLUSIONS 42

6.1 Main findings and conclusions 42
6.2 Limitations and future research 43

REFERENCES 45

APPENDICES 49

APPENDIX A - INTERVIEW GUIDES (SWEDISH) 49
A1 Interview guide: CH industry / Technical manager *49*
A2 Interview guide: Svenska Skydd *52*
A3 Interview guide: KFast *54*
APPENDIX B - GIOIA-INSPIRED CODING 57
Appendix B1: Initial Gioia Coding *57*
Appendix B2: Merge Groups and Theme Development *66*
Appendix B3: Development of Second-Order Themes and Aggregate Dimensions *71*
APPENDIX C - PHOTOS FROM PRODUCTION WALKTHROUGH. 73

ABBREVIATIONS

10R	Ten-resource hierarchy / 10R hierarchy
BIM	Building Information Modelling
CE	Conformité Européenne / European conformity marking
DFX	Design for X
EPD	Environmental Product Declaration
EU	European Union
QR	Quick Response
RFID	Radio-Frequency Identification
RISE	Research Institutes of Sweden

1. Introduction

This chapter presents the research area and explains the relevance of circularity in steel safety door manufacturing. It presents the background of the study, including circular economy, circular product design, and the role of design requirements in long-life industrial products. It also presents the problem statement, the aim and research questions, and the limitations of the study.

1.1 Background

Circular economy has become an increasingly important concept in sustainability research, European policy and industrial product development. It is commonly described as an alternative to linear product flows, where products are produced, used and discarded. Instead, circular economy aims to retain the value of products, components and materials for as long as possible by supporting strategies such as maintenance, repair, reuse, refurbishment, remanufacturing and recycling (Geissdoerfer et al., 2017; Kirchherr et al., 2017; Reike et al., 2018).

From a product development perspective, circularity is strongly influenced by design decisions. Bocken et al. (2016) explain that circular design strategies can contribute to slowing, narrowing, and closing resource loops. In this context, slowing resource loops means extending the useful life of products through durability, maintenance and repair, while closing resource loops means recovering materials so that they can reenter production systems. This makes design an important starting point for circularity, since decisions made during development can enable or limit later possibilities for maintenance, repair, reuse, refurbishment, remanufacturing and material recovery.

Design requirements are also central in this context because they translate product needs, functions and constraints into criteria that guide product development decisions (Dreier et al., 2025). In circular product development, such requirements need to consider not only technical performance and manufacturing conditions, but also later life-cycle considerations. Recent research shows that circularity design criteria need to be translated into more specific and measurable requirements in order to be usable in engineering practice (Dreier et al., 2025). This is particularly relevant for steel safety doors, where circular design requirements must be balanced against safety, certification, and production constraints.

The construction sector is especially relevant in this transition because it uses large amounts of resources and generates substantial waste. According to the European Commission, the built environment accounts for about 50% of all extracted material, while the construction sector is responsible for more than 35% of the EU's total waste generation (European Commission, n.d.-a). Construction and demolition waste also accounts for more than one third of all waste generated in the EU and includes material streams such as concrete, wood, glass, metals and plastics (European Commission, n.d.-b). This means that circularity in construction is not only a question of recycling materials at the end of life, but also about designing building products and components so that their functional and material value can be retained for longer.

Recent European regulatory developments further strengthen the relevance of circular product design and product information. The Ecodesign for Sustainable Products Regulation, which took effect in 2024, aims to improve the sustainability of products placed on the EU market by addressing aspects such as durability, reusability, upgradability, reparability, maintenance, refurbishment, remanufacturing and recycling (European Commission, 2024). The regulation also introduces the Digital Product Passport as a digital information carrier for products' components and materials. This passport can include information about technical performance, material origin, repair activities, recycling possibilities and environmental impacts. These developments show a broader movement toward product design that combines circularity with accessible and reliable product information.

Steel safety doors are a relevant product category within this broader construction and circular economy context. They are a long-life construction product used in public, commercial and industrial buildings where they contribute to fire safety, security, durability and building performance. As construction products, they are also connected to formal product compliance. Boverket explains that construction products covered by a harmonised standard or European technical assessment must have a declaration of performance and CE marking when placed on the European internal market, and that the declaration of performance carries product information through the distribution chain to the end customer (Boverket, 2021). Boverket also states that technical documentation should be traceable to the product from manufacturer to end user, which makes traceability relevant in the construction product context (Boverket, 2019).

For fire-rated doors, certification and testing are particularly important because the door is not assessed as an isolated door leaf. Boverket states that doors and openable windows are tested and classified as a complete unit, including the frame and accessories and that the fire classification is only valid for the tested and approved combinations (Boverket, 2024). RISE similarly describes fire resistance testing of doors, windows, gates, shutters and hatches according to relevant European standards, where the test report can be used as a basis for classification and CE marking (RISE, n.d.). Steel safety doors therefore represent a product area where circular economy ambitions, construction sector resource challenges, product information requirements and strict safety-related performance demands meet.

1.2 Problem Statement

Although circular design literature offers strategies for retaining product and material value, these strategies are often too general to provide direct guidance for design decisions in specific product contexts. The challenge is therefore to translate broad circular economy principles into product-specific and actionable design and traceability requirements for certified steel safety doors (Bocken et al., 2016; Dreier et al., 2025; Reike et al., 2018).

For certified steel safety doors, this challenge is particularly complex because circular interventions may affect the approved configuration and verified performance of the complete doorset. Fire-rated doors are tested and classified as complete units, including the door leaf, frame and accessories, which means that changes to materials, components or interfaces may require renewed assessment or certification (Boverket, 2024; RISE, n.d.). Circular design must therefore be developed within the boundaries set by safety, performance, and compliance requirements.

However, applying higher-value circular strategies such as repair, reuse, and refurbishment requires reliable product-specific information. These strategies generally retain more of the product's original function and value than material recycling (Reike et al., 2018). Such decisions require information about the door's classification and tested configuration, as well as its material composition, condition, use, performance, and life cycle. The relevant information must remain available and be updated throughout the product life cycle because changes in the door's condition, use, and performance can influence decisions concerning maintenance, repair, reuse, and refurbishment (Dreier et al., 2025; Markou et al., 2025). Without sufficient traceability, it is difficult to determine whether the door still corresponds to its tested and approved configuration and whether it is suitable for continued use, repair, reuse, or refurbishment (Boverket, 2024; Markou et al., 2025).

The problem addressed in this thesis is therefore that circular design strategies are available in theory, but there is limited guidance on how these strategies can be translated into actionable design and traceability requirements for certified steel safety doors. In this product context, circularity cannot be limited to end-of-life recycling. It must also address how repair, maintenance, disassembly, reuse, and refurbishment can be supported while maintaining certified safety performance and practical feasibility in industrial production. This creates a research gap between general circular design theory and the specific requirements of certified steel safety door systems.

1.3 Aim and Research questions

The aim of this thesis is to map the current design requirements and practices for steel safety doors in buildings and to develop recommendations for improving circularity in future steel door systems.

RQ1: What current design requirements and practices influence steel door design, and how do they support or limit circularity?

RQ2: How can circular strategies be translated into design recommendations for future steel safety doors, including end-of-life considerations, given the identified requirements and constraints?

1.4 Project limitations

This thesis is limited to a theoretical analysis and the development of concrete design proposals and recommendations for how circularity can be implemented into steel door design. The project does not include practical implementation, prototype development, or technical verification of the proposed changes.

In addition, the redesign of steel doors occurs relatively rarely, and models may remain in use for many years without major changes. Because of the industry's strict certification requirements, even minor design modifications would require new testing and re-certification, which falls outside the scope and time frame of this study. The results should therefore be regarded as strategic and conceptual recommendations that can support the company’s future product development.

2. Theoretical framework

This chapter presents the theoretical framework used in the analysis of the study's findings. It introduces circular economy, value retention, resource loops, the 10R hierarchy, Design for X, modularity, and traceability. These concepts are used to explain how circularity can be translated into design and traceability requirements while considering certification, product performance, and industrial production constraints.

2.1 Design for X

Design for X (DFX) refers to a family of design approaches where "X" represents a specific design objective, such as manufacturability, assembly, maintenance, disassembly, recycling or remanufacturing. The purpose of DFX is to make important product requirements explicit early in the product development process so that design decisions can better support later production and end-of-life activities (Kuo et al., 2001). In a circular economy context, DFX is relevant because it helps connect broad circular economy goals with more concrete product design strategies. Sassanelli et al. (2020) show that several DFX approaches are connected to circular economy, especially because they can support design decisions related to maintenance, recovery, recycling, remanufacturing, and end-of-life management.

In this thesis, DFX is used as a focused design perspective for translating circular economy principles into practical design considerations. The concept helps identify which product aspects should be considered during development in order to support later activities such as maintenance, disassembly, repair, recycling and remanufacturing. Rather than treating DFX as one single method, it is understood as a group of design approaches that can guide decisions depending on the chosen design objective (Kuo et al., 2001; Sassanelli et al., 2020; Mesa, 2023).

This means that circular design for steel safety doors should not only focus on environmental improvement in general but also consider which parts of the product can realistically be maintained, replaced, disassembled or recovered without compromising the certified performance of the complete doorset. From the circular design perspective, the most relevant DFX perspectives for this study are therefore connected to maintenance, component replacement, disassembly and remanufacturing. These areas support higher-value retention strategies such as repair, reuse and refurbishment rather than only material recycling at the end of life (Bocken et al., 2016; Reike et al., 2018). For this reason, DFX provides a useful way to structure circular design requirements in a product context where circularity must be balanced against performance requirements, production conditions, and certification boundaries. For this study, three DFX perspectives are especially relevant: Design for Disassembly, Design for Maintenance, and Design for Remanufacturing. These perspectives are used in this thesis to examine circular design in the context of steel safety doors.

2.1.1 Design for Disassembly

Design for disassembly refers to considering aspects such as reversible fasteners, appropriate disassembly sequences, and modular structures during product design to facilitate the separation of components and subassemblies. This approach can support more efficient disassembly throughout the product life cycle (Mesa, 2023). This is relevant because circular value retention depends on whether components and materials can be separated without unnecessary damage, excessive time, or complex tools. Formentini et al. (2025) emphasize that disassembly is important in circular product design because it affects several later life cycle activities including repair, reuse, refurbishment, remanufacturing and material recovery. Design for disassembly is therefore not only an end-of-life issue, but also a design condition that can influence whether higher-value circular strategies become practically possible (Formentini et al., 2025).

In the context of steel safety doors, design for disassembly does not mean that the whole product should be easy to take apart without restrictions. Since the certified performance of the door depends on the complete doorset, disassembly must be considered in relation to approved components, interfaces and safety requirements (Boverket, 2024). Relevant design guidelines considered in this study are therefore to use accessible and reversible connections where possible, reduce unnecessary permanent joining, support safe separation of material layers, and make components such as seals, locks, hinges, and fittings easier to remove and replace when this does not affect certification. These guidelines are also connected to modularity, since standardized and accessible interfaces can support component replacement and maintenance without requiring major changes to the whole product system (Fixson & Park, 2008; Formentini et al., 2025).

2.1.2 Design for Maintenance

Design for maintenance refers to considering maintenance activities during the product design process to facilitate further maintenance of the product (Vaneker & van Diepen, 2016). This approach is important because steel safety doors are long-life products that must continue to perform over many years. Circular strategies for slowing resource loops emphasize durability, maintenance and repair as ways to extend product lifetime and reduce premature replacement (Franconi et al., 2022; Jerome & Ljunggren, 2025). Repair can also be considered a higher-value circular strategy than recycling because it preserves more of the product's original function and value (Morseletto, 2020). For this reason, maintenance should be considered as a central part of circular design, rather than only as an activity that happens after installation.

For steel safety doors, relevant maintenance guidelines considered in this study include supporting inspection, servicing, and replacement of wear components during the use phase. This includes designing for access to parts that may need maintenance, such as seals, hinges, locks, closers and fittings. It also includes making product information available so that maintenance actors can understand which components are approved, what has been replaced, and whether the door still corresponds to its required classification. This links design for maintenance to traceability, since structured product information can support decisions about repair, reuse, refurbishment, and end-of-life handling (Markou et al., 2025).

2.1.3 Design for Remanufacturing

Design for remanufacturing refers to considering remanufacturing requirements during the product design process, with the aim of designing products that can be more effectively and efficiently remanufactured in further use cycles (Boorsma et al., 2021). Paul et al. (2024) describe remanufacturing as a process that can return used products to a functional state. From a circular economy perspective, remanufacturing can retain more product value than material recycling because more of the original product structure, material, and function are preserved (Singhal et al., 2020).

In the case of steel safety doors, remanufacturing is more complex than for many ordinary industrial products because the door must still meet safety and certification requirements after any intervention. Since fire-rated doors are tested and classified as complete units, replacing or restoring components may affect whether the original classification can still be verified (Boverket, 2024; Prieler et al., 2023). Therefore, relevant design guidelines considered for this study include controlled inspection, clear identification of components, documentation of product history, and defined procedures for which parts may be replaced or restored. Remanufacturing may not always be possible for certified steel safety doors. However, the perspective is still useful because it clarifies what information, interfaces and component choices would be needed if future reuse or refurbishment systems are to become more feasible (Ostapska et al., 2024).

Overall, DFX provides a useful way to connect circular economy theory with practical product design requirements. For this thesis, DFX helps identify guidelines relevant to steel safety doors: accessible and reversible connections, replaceable wear components, maintenance-friendly interfaces, clear product documentation, and design choices that support future disassembly, reuse, refurbishment, or material recovery (Mesa, 2023). These guidelines are relevant because circularity in steel safety doors needs to be considered in relation to certification, safety performance, and long service life.

2.2 Circular Design Strategies for Value Retention

Circular design strategies describe how product design can support circular economy goals by keeping products, components and materials in use and by reducing unnecessary resource loss. In this study, value retention refers to the preservation of functional and material value over a product's life cycle through strategies such as maintenance, repair, refurbishment, remanufacturing and recycling (Bocken et al., 2016; Reike et al., 2018).

Circular product design strategies should be considered early in product development because substantial changes become more difficult once key product specifications, resources, infrastructure and activities have already been committed to a particular design direction (Bocken et al., 2016). Circular design literature also emphasizes that broad circular economy principles need to be translated into more concrete design strategies that can guide product development practice. Moreno et al. (2016) highlight product life extension, multiple use cycles and recovery as important considerations in circular design, while Franconi et al. (2022) structure circular product design around objectives such as maintenance and longevity, reuse, refurbishment, remanufacturing and recycling. In this thesis, circular design strategies are used as a theoretical basis for examining how steel safety door design can support value retention through maintenance, repair, disassembly, reuse, remanufacturing, and material recovery while considering certification, performance, and production constraints. The following section therefore discusses circular strategies for slowing, narrowing, and closing resource loops.

2.2.1 Resource loops

Resource loop strategies differ in how they seek to retain value and reduce resource use. Slowing resource loops focuses on extending the useful life of products and components, while narrowing resource loops focuses on reducing the amount of material, energy, and other resources required per product or function (Bocken et al., 2016). Closing resource loops focuses on recovering components and material after use so that they can enter new production or material cycles instead of becoming waste (Geissdoerfer et al., 2017; Reike et al., 2018).

The DFX perspectives introduced in section 2.1 can support these three resource loop objectives in different ways. Strategies related to product life extension primarily support slowing loops, while strategies aimed at reducing material use and production losses support narrowing loops (Sassanelli et al., 2020; Mesa, 2023). Recovery-oriented strategies support closing loops by facilitating component recovery, material separation, remanufacturing, and recycling. Some DFX strategies, particularly design for disassembly, can contribute to more than one resource loop objective because they can support both component-level interventions during use and recovery at the end of life (Bocken et al., 2016; Formentini et al., 2025). Table 1 presents the relationship between the three resource loops objectives and the corresponding DFX strategies.

Table 1. Design strategies for slowing, narrowing, and closing resource loops

Resource-loop strategy	Circular objective	Relevant design strategies
Slowing resource loops	Extend the useful life of products and components	Design for maintenance; Design for disassembly; durability and repair-oriented design
Narrowing resource loops	Reduce material and resource use per product or function	Reduced material use; material efficiency; reduction of production losses
Closing resource loops	Recover components and materials for new cycles	Design for Disassembly; Design for Remanufacturing; material separation and recycling

In this study, slowing resource loops is particularly relevant because steel safety doors are long-life products, and several of the identified circular opportunities concern maintaining their function and avoiding premature replacement. Narrowing strategies are considered where material and production efficiency can be improved without compromising required product performance, while closing strategies become relevant when continued product or component use is no longer feasible (Bocken et al., 2016; Dreier et al., 2025). Although Table 1 presents several design strategies associated with the three resource loops objectives, the DFX perspectives examined in more detail in this study are Design for Maintenance, Design for Disassembly and Design for Remanufacturing, as these are considered most relevant to the circular design of steel safety doors. These DFX perspectives can support circular product development by facilitating activities such as maintenance, component recovery, remanufacturing, and end-of-life management (Sassanelli et al., 2020).

2.3 10R Hierarchy

Several R-frameworks have been developed to structure circular economy strategies. These frameworks commonly arrange strategies in a hierarchy according to their level of circularity (Dreier et al., 2025; Potting et al., 2017). Earlier frameworks often focused on Reduce, Reuse, and Recycle, but later frameworks have added more strategies such as Repair, Refurbish, Remanufacture, and Recover (Dreier et al., 2025). More than 55 R-terms have been identified within the circular economy framework, extending far beyond Reduce, Reuse and Recycle. This large number of terms indicates a need for clearer structure (Zorpas, 2024). Therefore, several R-frameworks have been developed with the aim of classifying circular strategies and assessing their level of circularity. These frameworks range from more concise 3R approaches to more detailed 9R and 10R hierarchies (Dreier et al., 2025; Muñoz et al., 2024; Reike et al., 2018).

Within this literature, Reike et al. (2018) present a 10R typology of circular strategies. The typology synthesizes existing R-strategy perspectives into resource value retention options that rank circular strategies from R0 to R9 according to their ability to preserve product and material value. These resource value retention options are further categorized into short loops, medium loops, and long loops. The loop length reflects how closely a product remains to its user and original function across circular strategies, with longer loops involving more processing and often a loss of original function (Reike et al., 2018).

Table 2. Overview of the 10R hierarchy and corresponding loop categories (adapted from Reike et al., 2018).

Short loop	R0	Refuse	Avoid the use of a product, material, or unnecessary packaging altogether.
	R1	Reduce	Minimize material and energy use, as well as waste generation, across the product life cycle.
	R2	Reuse	Enable the continued use of a product by another user with little or no modification.
	R3	Repair	Restore a product to working conditions by fixing or replacing faulty components without significantly changing its design or function.
Medium loop	R4	Refurbish	Upgrade a product by repairing or replacing key components while retaining its overall structure.
	R5	Remanufacture	Fully disassemble, inspect, clean, and restore a product to its original or a like-new condition.
	R6	Repurpose	Adapt a discarded product or component for a different function.
Long loop	R7	Recycle	Process discarded products or materials into secondary raw materials.
	R8	Recover	Recover energy or remaining value from waste streams, typically through energy recovery processes.
	R9	Re-mine	Retrieve materials or valuable parts from landfills or waste deposits after disposal.

More recent research has developed the R-hierarchy further by translating it into circularity assessments. Muñoz et al. (2024) connect the R-hierarchy to circularity assessment by showing how indicators and indexes can be used to measure circular economy practices. Circularity assessment in this context can be understood as the use of indicators, indexes, or scoring methods to evaluate circularity in products and material flows (Muñoz et al., 2024). The authors argue that existing assessment indexes do not fully reflect the priority order of the 10R strategies. This is important because the 10R hierarchy gives higher priority to strategies that retain more value. Recycling and recovery can recover material or energy value, but they break down the product as a functional whole and therefore retain less product integrity than reuse, repair, refurbishment, and remanufacturing (Morseletto, 2020). To address this limitation, Muñoz et al. (2024) develop a 9R circularity index that translates the R-hierarchy into a quantitative scoring system for comparing circular and linear models. However, because circularity assessments do not explain how products should be designed and because the product-design implications of R-strategies are not always clearly defined, circular design criteria need to be translated into measurable product requirements (Dreier et al., 2025).

The 10R hierarchy provides a structure for distinguishing between circular strategies that retain product value and strategies that mainly recover material or energy value (Reike et al., 2018). For steel door manufacturing, circularity should not only be understood as the recyclability of materials. Steel safety doors mainly consist of recyclable materials and components, but higher-value retention strategies should be considered first. Recycling and recovery can recover material or energy value, but these strategies do not retain the product's original form or function (Morseletto, 2020). The hierarchy can therefore clarify which strategies are preferable (Reike et al., 2018). However, it does not explain how these strategies can be enabled through product design (Dreier et al., 2025). For this reason, the 10R hierarchy is used in this thesis to guide the selection of strategies that retain more value whenever feasible. The other sections focus on how these strategies can be enabled through modularity, traceability, and Design for X.

2.4 Modularity in industrial products

According to Ulrich (1995), product architecture is defined as "the scheme by which the function of a product is allocated to physical components" and consists of three elements: the arrangement of the product's functional elements, the mapping of those functional elements to physical components, and the specification of the interfaces between interacting components. The arrangement of functional elements describes what the product is intended to do. The mapping of functions to components specifies which function each component implements and whether those functions are shared or distributed. This could mean that one component implements multiple functions, or that one function is realized by multiple components. Finally, interface specification defines how components interact and how components exchange loads, energy, or information (Ulrich, 1995). Modularity within industrial products is closely linked to the concept of product architecture. A modular product architecture is characterized by relatively independent components connected through standardized and decoupled interfaces, whereas an integral architecture exhibits stronger coupling between components and functions (Fixson & Park, 2008).

The characteristics of component interfaces play a central role in determining the degree of modularity in industrial products. Interface properties such as standardization, reversibility, and accessibility influence how easily components can be removed or replaced during the product's lifetime. Modular architectures with well-defined and decoupled interfaces can reduce dependency between components, thereby enabling maintenance and component-level interventions without affecting the integrity of the overall system (Fixson & Park, 2008).

There are several design strategies through which modularity can be achieved. Standardized interfaces are emphasized as a way to enable substitution of component variants without redesigning other components (Sanchez & Mahoney, 1996). In addition to interface design, modularity is shaped by how functions are allocated to components, together with interface characteristics such as how parts connect and how easily they are separated (Fixson & Park, 2008). From a circular design perspective, modularity is closely related to design-for-disassembly principles, including the use of mechanical and reversible connectors (Ostapska et al., 2024), and to repair-oriented design choices that support tool access, component accessibility, and reversible connections (Formentini et al., 2025).

For steel safety doors, modularity is relevant since the doors consist of several components with different functions and replacement requirements. Some components are closely linked to the tested fire performance and classification of the complete door assembly, which can limit possible design changes (Boverket, 2024; RISE, n.d.). Other components can provide greater opportunities for maintenance, replacement, and disassembly. A modular perspective can therefore provide an understanding of how the allocation of functions, and the design of component interfaces affect the possibilities for circular design. In this thesis, modularity provides a useful perspective for examining how circular design strategies can be applied to steel safety doors while considering practical and regulatory constraints.

2.5 Traceability

Traceability refers to the ability to connect a physical product or component to reliable information about its material, components, classification, and life-cycle history. In circular design, this information is important because decisions about maintenance, repair, reuse, refurbishment, and recycling depend on knowing what the product contains and how it has been used (Jensen et al., 2023; Çetin et al., 2023; Markou et al., 2025).

In practice, traceability depends on two things: first, a stable way to identify the physical product or component, and second, a digital information structure that makes the stored data usable and transferable across the life cycle. The review also points out that many passport approaches are still largely static and rely on manual updates, which increases the risk that information becomes outdated over long use phases. To make traceability more actionable, the paper discusses a shift toward more dynamic passports with more frequent updates, for example, by connecting BIM (Building Information Modelling), a shared digital information model used to structure and manage asset/product data, to physical carriers such as QR codes or RFID (Radio-Frequency Identification) tags, so that the product and its data remain linked over time (Markou et al., 2025).

For construction products, traceability is also connected to compliance and product documentation (Boverket, 2019). Boverket highlights the role of the declaration of performance and CE marking in carrying product information to the customer (Boverket, 2021). Recent EU policy further strengthens this through the digital product passport, which aims to make product-related information more accessible across the value chain (European Commission, 2024).

Traceability is relevant in circular design because it supports access to product-specific information needed for repair, refurbishment, reuse, and take-back decisions. By making information about materials, components, and changes during the use phase more accessible, traceability can reduce uncertainty and support more informed circular decision making across the product life cycle (Markou et al., 2025).

For steel safety doors, traceability is especially important because reuse or component replacement cannot be judged only from the physical condition of the door. Information about fire classification, approved component combinations, maintenance history, and material content is needed to assess whether circular strategies can be applied without compromising verified performance (Jensen et al., 2023). Traceability can therefore support circularity for steel safety doors by linking product information to maintenance, disassembly, reuse, and end-of-life handling.

3. Research method

This chapter outlines the methodology used in the thesis. It presents the abductive approach, the qualitative single case study design, and the case context. It then describes the primary and secondary data collection, including the benchmarking process, followed by the data analysis. Finally, the chapter presents a discussion of research quality and the limitations of the study.

3.1 Methodological approach

In this thesis, an abductive approach is used. This means working iteratively between theory and empirical data, rather than moving in only one direction from theory to data or from data to theory. Existing theory was used to frame the study and to identify relevant concepts related to circularity, while empirical observations were used to assess how these concepts could be understood in this specific case. As the understanding developed, the interpretations of the empirical material were continuously adjusted through comparison between the theoretical framework and the empirical material. In this way, theory and observation informed each other throughout the study (Säfsten & Gustavsson, 2020).

This abductive approach is suitable for this thesis because the study involved an iterative process between circular economy literature and case-specific material from CH Industry, Svenska Skydd and KFast. Svenska Skydd is a company that supplies and installs security doors and provides stairwell renovation services, primarily for housing companies, property owners and housing associations, and is part of the same corporate group as CH Industry (Svenska Skydd, n.d.). KFast, formally known as Eskilstuna Kommunfastigheter AB, is a municipally owned property company that owns, manages and develops residential properties and municipal premises in Eskilstuna (KFast, n.d.). The empirical data mainly consisted of notes from meetings with CH Industry, along with notes and photographs from the production walkthrough and notes from the meeting with Svenska Skydd. The interviews and meeting data were analyzed using a Gioia-inspired approach, based on the structure of first-order concepts, second-order themes and aggregate dimensions (Magnani & Gioia, 2023). Circular strategies and concepts identified in the literature review were used as an initial analytical starting point, but empirical findings were also used to refine the focus of the study and to guide further review of relevant literature. This made it possible to identify circular strategies that were not only theoretically relevant but also realistic in relation to the industrial conditions of CH Industry's steel safety doors.

3.2 Research design

This study uses a qualitative single case study design. A qualitative approach is suitable because the thesis examines current practices, requirements, and improvement needs related to CH Industry's steel safety doors in their empirical setting. The purpose is not to measure variables or test statistical relationships, but to understand how circularity, certification, documentation, and product development are interpreted and handled by actors connected to the product's value chain (Säfsten & Gustavsson, 2020).

The single case study design is suitable because the thesis focuses on one specific industrial case, CH Industry's steel safety doors. The aim is to develop an in-depth understanding of one product and company rather than to compare several companies. This is appropriate because this study investigates how circular strategies can be understood and translated into design and

traceability requirements within a certified steel door context. Case study research is relevant when the aim is to study a phenomenon in its real-life context and to connect the research design, data collection and analysis to the specific case (Yin, 2018).
In this study, “what” questions support the mapping of requirements and current practices, while “how” questions support the development of guiding knowledge and recommendations for action. Although this single case design does not aim for statistical generalization, it can support analytical generalization by developing contextually grounded insights that may be relevant for similar industrial product contexts (Yin, 2018).

The case is centered on CH Industry’s steel safety doors, but it also includes perspectives from actors connected to the product’s value chain. Svenska Skydd is included because of its role in sales, installation and replacement of steel doors. KFast is included in providing a property owner and maintenance perspective on reuse, documentation, classification issues, and practical barriers in renovation and replacement projects. These actors were included because they provide empirical perspectives that are relevant to the same product context, while the study remains focused on one single case.

The study also combines several types of empirical material, including meetings, interviews, observations, written notes, and photographic documentation. This supports the case and strengthens the empirical basis through triangulation where different data sources are used to complement each other (Säfsten & Gustavsson, 2020).

3.3 Case description

This thesis was conducted as a case study in collaboration with CH Industry, which is an Eskilstuna-based manufacturing company with experience in industrial production and contract manufacturing. The company was founded in 1948 and specializes in contract manufacturing, primarily producing industrial components for heavy-duty vehicles and off-road machinery. Since 2018, CH Industry has also operated as a product-owning company specializing in fire and security doors (CH Industry, n.d.).

The case in this study is centered on CH Industry’s steel door products and was defined in collaboration with the company, based on its interest in exploring how circular strategies could be introduced into future steel door products. The study also includes empirical input from Svenska Skydd, which is responsible for sales and installation of the doors and KFast, which contributes a property owner and maintenance perspective on reuse and circular handling in renovation and replacement projects. This is relevant because the company provides practical insight into door replacement, reuse potential, renovation conditions and end-of-life handling.

Although the study focuses on one specific case, CH Industry is still relevant to the broader steel-door sector because it operates within the same type of certified product context in which safety requirements, production conditions and downstream handling influence the possibilities for circular design.

3.4 Data collection

The data collection in this thesis consists of both primary and secondary data. The primary data consists of notes from meetings, observation notes, photographs from the production walkthrough, notes from the meeting with Svenska Skydd and the semi-structured interview with KFast representatives. The secondary data consists of academic literature, standards, certifications, related sources and observed product labels. Using both primary and secondary data makes it possible to combine different types of information and strengthen the empirical basis of the study through triangulation (Säfsten & Gustavsson, 2020).

3.4.1 Primary data collection

Table 3. Overview of primary data collection

Data source	Role/position	Meeting time	Format	Main empirical contribution	Relevance to the study
CH Industry	Former company representative	1 h 30 min	on-site meeting	Provided insight into the manufacturing environment, visible door components, production flow, labels, joining methods and conditions relevant to disassembly and traceability.	Supported understanding of how steel safety doors are currently produced, and which production-related conditions may influence circular design possibilities.
Svenska Skydd	Sales and installation representative	15 min	Semi-structured interview	Provided practical insight into installation, replacement, removal of existing doors, reuse possibilities and handling of removed steel doors.	Contributed a downstream perspective on how steel safety doors are handled after production and during replacement situations.
KFast	Property management & Reuse and maintenance perspective	33 min	Semi-structured interview	Provided insight into reuse in renovation projects, including inventories, internal storage, documentation needs, classification issues, hazardous material, and barriers related to time, cost, and knowledge.	Contributed a property owner and downstream perspective on the conditions that affect reuse, recovery and circular handling of steel doors after installation and during renovation or replacement projects.
CH Industry	Technical manager	32 min	Semi-structured interview	Provided detailed insight into CH Industry's current steel safety door design, including fire and certification requirements, material selection, standardized product architecture, joining methods, and constraints relevant to circular design.	Contributed to the understanding of how certification requirements, material selection, standardized construction, and joining methods influence circularity in steel safety doors

Primary data was generated through direct contact with the case context.

The empirical data collection started with the initial on-site meeting at CH Industry. During this meeting the project scope and practical conditions for the thesis were discussed. As part of the same visit, a guided walkthrough of the production floor was carried out. The walkthrough was used to build an initial understanding of the manufacturing environment and to document production-related conditions relevant to circular strategy implementation. During the walkthrough, photographs were the primary source of data capture and were supplemented by notes based on observations and explanations given during the visit.

The observations during the walkthrough focused on product-related and production-related aspects considered relevant to circularity such as visible door components, labels, connections, material combinations, manufacturing flow and conditions that could influence disassembly, replacement and traceability. The observation process was exploratory but guided by the aim of the thesis and by the concepts identified in the literature review. Notes were taken during the visit in order to capture both what was observed and how this was explained by Daniel at CH Industry. The photographs documented examples of doors, components, labels and production-related details that were considered relevant for later analysis. In this way the walkthrough functioned both as an observation and a way of supporting the later analysis through empirical documentation.

During the initial meeting with CH Industry, it became apparent that contact with Svenska Skydd would be valuable in order to gain a better understanding of installation, replacement and removal of existing doors. A meeting with Svenska Skydd was therefore arranged. This meeting was conducted as a semi-structured interview where prepared questions were used as guidance while still allowing follow-up questions and clarifications. This type of data collection is suitable when the aim is to gather practical and experience-based information while still maintaining some structure in relation to the purpose of the study (Säfsten & Gustavsson, 2020).

In addition to the interview with Svenska Skydd, a semi-structured interview was conducted with David Diaz, Technical Manager at CH Industry in order to gain a deeper understanding of door construction, material selection, joining methods and how certification requirements influence possibilities for circular design. This interview focused on the technical design of the door, current assembly solutions, material related constraints and potential opportunities for repair, reuse and future design changes. Notes from the interview were used as empirical material in the analysis and in the findings related to door construction and circularity.

In addition to the interviews with Svenska Skydd and CH Industry, a semi-structured interview was conducted with KFast representatives in order to gain a broader property-owner and maintenance perspective on reuse and circular handling of steel doors. The interview focused on how reuse is handled in property and renovation projects, including reuse inventories, internal storage, CCBuild, handling of removed doors, hazardous substances, classification issues, documentation needs and practical barriers related to time, cost and knowledge. This interview was relevant because it provided insight into the conditions that affect reuse, recovery and circular handling after installation and during renovation or replacement projects.

The interview guides were prepared based on the aim of the thesis and on themes identified in the literature review and the initial case contact. The questions focused on topics relevant to each interview, such as installation practice, replacement situations, dismantling conditions, reuse possibilities, handling of removed doors, material related issues, door construction, joining methods and certification-related constraints. This made it possible to keep the

discussion focused on the purpose of the study while still allowing the respondents to elaborate on practical examples and clarify work procedures when relevant. Notes were taken during the meeting and were compiled afterwards in order to preserve the empirical content in a structured way. The interviews were conducted through Microsoft Teams, where the meetings were recorded and automatically transcribed. The transcripts were then reviewed and corrected manually before being used as empirical material in the analysis.

The selection of empirical sources was made purposefully based on the aim of the thesis. CH Industry was selected as the main case company since the study focuses on circularity in relation to its steel safety door production. Svenska Skydd was included because the company contributes a practical downstream perspective on manufacturing and field level handling within the same case context. KFast was included to add a property owner and maintenance perspective on reuse, documentation, classification issues and practical barriers in renovation and replacement projects within the same single case study design (Säfsten & Gustavsson, 2020).

3.4.2 Secondary data collection

The distinction between primary and secondary data was made based on whether the material was generated directly during the project or whether it consisted of already existing sources. The secondary data consisted of published and preexisting material that was relevant for understanding circular design, certification requirements and traceability in relation to steel safety doors.

The literature review was carried out in order to build the theoretical foundation of the thesis. The review focused on literature related to circular economy, value retention strategies, traceability, modularity and design for X. Since steel safety doors are strongly influenced by certification and regulatory requirements, standards and certification related sources were also included. The literature review was not only conducted at the beginning of the study but continuously during the study. This is consistent with the abductive approach where theory and empirical material are considered iteratively and where the understanding develops step by step during the project (Säfsten & Gustavsson, 2020).

The literature review had a mainly narrative and exploratory character, since the purpose was to identify concepts, design strategies and requirements areas relevant to the case rather than to systematically review all available research in one narrowly defined field. The literature search was carried out through academic search tools and university library resources using combinations of keywords related to the aim of the thesis. Examples of keywords used in the search process were circular economy, circular design, value retention, 10R, repair, reuse, remanufacturing, modularity, traceability, material passport, design for X, steel doors, safety doors, fire doors and certification. The selection of literature was based on relevance to the research questions and to the industrial case. Priority was given to peer-reviewed journal articles, standards, official sources and certification related material that could support the understanding of both circular design and the specific constraints related to certified steel safety doors.

3.4.2.1 Benchmarking

Benchmarking was used to identify and compare circularity practices relevant to steel safety doors, both within the steel door sector and in related product categories. The purpose was to examine how different companies work with circularity and what practices support improvement. This benchmarking approach can be described as process benchmarking, since it focuses on how companies communicate and support circularity practices through publicly available documents, rather than on a single set of performance indicators (Bhutta & Huq, 1999).

Data were collected from company websites, product information, and environmental product declarations. Selected cases included Daloc, Teckentrup, Novoferm, ASSA ABLOY and ReDoor. For each case, information related to circularity was gathered and reviewed, including claims and evidence concerning recyclability, design for disassembly, maintenance, adaptability, reuse, and environmental documentation.

Based on the collected data, a benchmark table was developed to compare selected companies and cases in a structured way. The table presents each company or case together with its country and evaluates them based on relevance to steel safety doors, main circularity evidence, evidence strength, and overall benchmark value. The relevance level reflects how closely each case matches the product focus of steel safety doors, whereas evidence strength reflects how well the identified circularity claims are supported by specific and verifiable sources. The complete benchmark table is presented in the Results chapter.

3.5 Data analysis

The interview and meeting data were analyzed using an approach inspired by the Gioia methodology. This method provided a clear structure for qualitative analysis through first-order concepts, second-order themes, and aggregate dimensions. It also aligned with the study's abductive design, since the analysis developed through ongoing comparison between empirical material and theory (Magnani & Gioia, 2023).

All interviews took place in Microsoft Teams and were recorded there. Teams generated automatic transcripts during each recording. These transcripts were then reviewed and edited by hand to correct errors. The revised transcripts formed the basis of the analysis.

Following the logic of the Gioia methodology, the analysis began with a first-order coding process in which concepts were kept close to the respondent's own wording and descriptions. This initial coding generated a broad set of first-order concepts grounded in the empirical material rather than in predefined theoretical categories. In practice, each transcript was read multiple times to identify excerpts relevant to the research questions. For each relevant excerpt, a condensed interpretation was written in order to clarify its meaning, after which a tentative first-order concept was assigned.

Next, the first-order concepts were compared across interviews and meetings to identify similarities, differences, and recurring patterns. Through this iterative process, related concepts were grouped into broader second-order themes. As part of this step, similar first-order concepts were merged in order to reduce overlap and make patterns across respondents more visible. These themes were then condensed into aggregate dimensions that captured higher-level patterns in the empirical material. In this way, the analysis moved from descriptions close

to the respondent's accounts to more abstract analytical categories. The resulting structure was then organized into a Gioia-inspired data structure in order to make the progression from empirical material to analytical interpretation more transparent.

The development of second-order themes and aggregate dimensions was carried out abductively through continuous comparison between empirical material and the theoretical framework. More specifically, the emerging themes were interpreted in relation to concepts such as value retention, modularity, traceability, and circular design requirements. This made it possible to build a clear connection between empirical material and theoretical interpretation, which is central to the Gioia methodology. The purpose of the analysis was therefore not statistical generalization, but a structured and theoretically informed interpretation of the case.

3.6 Research quality

Since this thesis is based on qualitative single case study research, research quality is discussed mainly in terms of trustworthiness rather than statistical generalization. Trustworthiness was strengthened by using several relevant data sources and by keeping the interpretation closely connected to both the empirical material and the theoretical framework. This increased the credibility of the analysis and made the reasoning more transparent.

Source criticism was important, especially in relation to secondary data such as academic literature, standards and certification related sources. These sources were selected because they are relevant to the case, but they were interpreted critically and in relation to the purpose of the study. Particular attention was given to whether the sources were academically reliable, officially established, or directly relevant to steel safety doors and circular design.

Ethical considerations in this thesis mainly concern responsible use of company related information and fair representation of the participating actors. A limitation of the study is that it is based on one main case company and a limited number of downstream and property management actors, mainly represented by Svenska Skydd and KFast. The study also does not include prototype development, technical testing, or validation of the proposed recommendations. The results should therefore be understood as analytically grounded recommendations rather than technically verified solutions.

Taken together, the use of several data sources, step by step documentation of the analysis and the close connection between empirical material and theory contributed to the trustworthiness of the study. Credibility was strengthened by combining interviews, observations, photographs and secondary material while dependability was supported by documenting how the empirical material was reviewed, coded and interpreted. However, since the study is based on one main case company and a limited number of downstream and property management actors, the findings should be understood as contextually grounded rather than statistically generalizable.

4. Results

This chapter presents the empirical findings of the study. More specifically, it presents findings from the interviews, observations related to certification and performance requirements and the benchmarking of circularity in the steel safety door sector.

4.1 Benchmarking Results

The benchmarking shows that companies in the steel door sector mainly communicate circularity through recyclability, environmental documentation, long service life, and limited forms of product adaptation or reuse. Across the reviewed companies, the most common circularity-related claim concerned steel as a recyclable material. Since steel is the main material in these products, several companies present their doors as circular on the basis of material recyclability. At the same time, the benchmarking revealed fewer examples of strategies aimed at preserving the value of the complete product or its components over time.

The reviewed companies differed in how they addressed circularity. Daloc presented the widest range of circularity-related practices among the benchmarked companies. These included recyclability, maintenance, disassembly, and limited reuse practices. Daloc also offers reused doors, although these are sold without classification. Reuse was identified in the sector, but only in limited cases and with constraints related to classification and verified performance (Daloc AB, 2025).

The ReDoor project was identified as an example of ongoing work on reuse in safety related doors. The project focuses on the possibility of reusing doors while maintaining verified fire safety performance. The project addresses reuse in relation to certification and verified fire safety performance (DBI, 2025).

Teckentrup and ASSA ABLOY mainly addressed circularity through environmental product declarations. These documents provide information on materials, life cycle impacts, and end-of-life scenarios. The benchmarking therefore shows that environmental documentation is an important way of communicating circularity in the sector. The reviewed documentation included information on materials, life cycle impacts, and end-of-life scenarios, while fewer examples were found of product-level strategies such as reuse, repair, or refurbishment (ASSA ABLOY Security Solutions Metal Industries Factory [LLC], 2019; Teckentrup GmbH & Co. KG, 2012).

Novoferm presented adaptability as part of its circularity approach. This was mainly expressed through retrofittable components and standardized fittings. These features support maintenance, component replacement and upgrade. Novoferm focused more on interventions during the use phase, while the other companies focused mainly on recyclability (Novoferm Group, n.d.).

The benchmarking results summarized in Table 4 suggest that circularity in the steel door sector is still limited and is mainly expressed through recyclability, environmental transparency, and some examples of reuse or adaptability.

Table 4: Summary of circularity approaches identified in the benchmarking

Company	Country	Relevance	Main circularity evidence	Evidence strength	Benchmark value
Daloc	Sweden	High	Design for disassembly; 97% recyclable; recycled steel option; reuse practices	High	Strong direct circularity benchmark
ASSA ABLOY / Prometal	UAE / Middle East	High	Product-level EPD; reuse during service life; recyclable packaging; end-of-life recycling routes	High	Strong product-data benchmark
Novoferm	Germany / Europe	Medium	100% recyclable steel profile; retrofittable components; standardized fittings	Medium	Strong adaptability benchmark
Teckentrup	Germany	High	EPD for steel multifunction doors; cradle-to-grave life cycle data;	Medium	Strong transparency benchmark
ReDoor	Denmark	Medium	Reuse of fire doors; performance verification in reuse context	High	Strong reuse-innovation benchmark

4.2 Regulatory and production constraints

The empirical material showed that current steel door systems are strongly shaped by regulatory, performance, and production-related constraints. As shown in Figure 1, this section includes requirements related to fire performance, certification and approval procedures, standardized door architecture, and current joining and assembly solutions. Across the interviews, respondents described how these constraints influence both the design of current steel doors and the possibilities for redesign, reuse, and end-of-life handling.

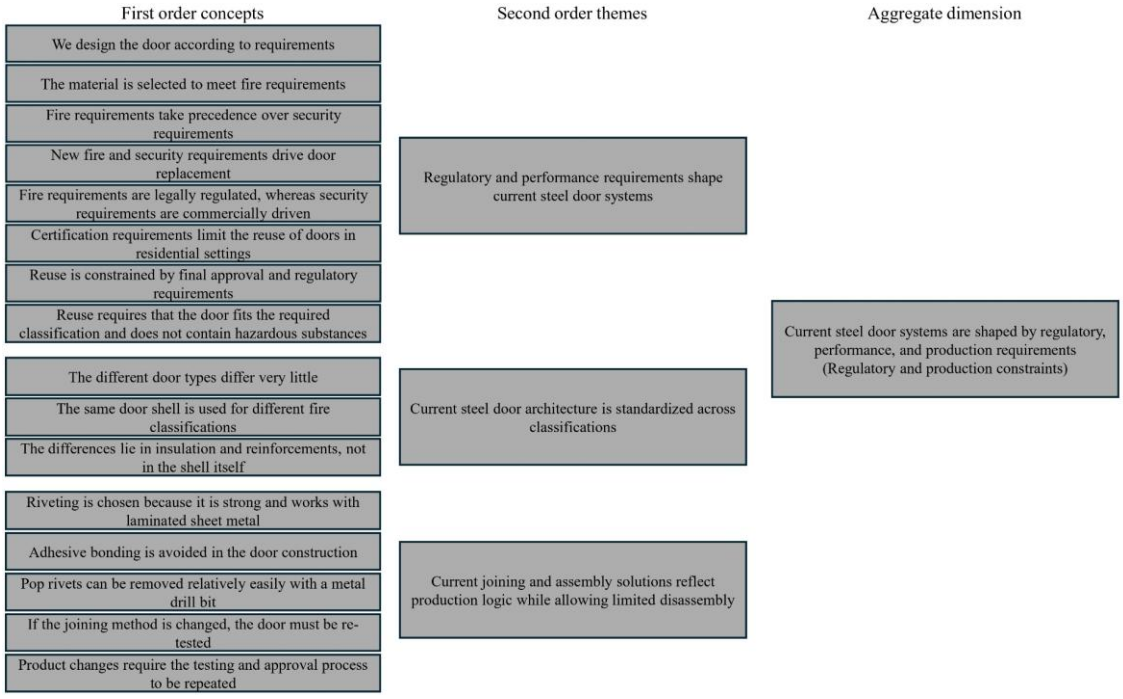


Figure 1: Regulatory and production constraints in the Gioia-inspired data structure.

A recurring finding across the interviews was that current steel door design is governed by regulatory and performance requirements. The technical manager at the steel door manufacturer explained that, in the design process, doors are developed according to specified requirements and continuously adjusted when new requirements are introduced. As he described it, the door design is developed *"according to the requirements"* and is revised when new requirements are added. Fire-related requirements were described as legally regulated and therefore more difficult to compromise than other requirements. One respondent explicitly distinguished fire requirements from security requirements by explaining that fire classification is *"a legally regulated requirement,"* whereas security classification is commercially driven. Similar patterns were visible from the installer perspective, where old doors were mainly replaced because they no longer met current requirements related to fire, gas tightness, sound insulation, and security. From the municipal and reuse-oriented perspective, approval procedures and required classifications were also described as important factors that affect whether reused doors can be accepted in a new context. Taken together, these findings show that current steel door systems are not primarily shaped by circularity goals, but by the need to satisfy formal and performance-based requirements.

The empirical material also showed that current steel door architecture is highly standardized across classifications. According to the technical manager, different door classes are built on the same general shell, while differences between classifications are mainly achieved through

internal adjustments such as insulation and reinforcements. As he explained, the company has developed *"the same shell on the door"* for different classifications, meaning that a 15-, 30-, or 60-minute fire-rated door is visually the same door. In this sense, the visible outer structure remains largely unchanged across different fire ratings, while performance is varied through internal components. This indicates that standardization is an important part of the present design logic. Rather than redesigning each door class from scratch, the system relies on a shared basic structure that can be adapted internally to meet specific classifications. The findings therefore suggest that current steel door systems combine standardization with requirement fulfilment, which may support efficient production but also reinforce existing design logic.

A third result within this aggregate dimension concerns current joining and assembly solutions. The steel door manufacturer described riveting as a strong and practical joining method that works well with laminated sheet metal, whereas welding was described as unsuitable because the heat could damage the surface treatment. The respondent described riveting as *"one of the strongest ways to join two sheet metal pieces,"* while also explaining that the laminated sheets cannot easily be welded. Glue was also largely avoided in the construction. These accounts show how current joining methods are closely connected to the chosen materials and the production process. At the same time, the findings suggest that current assembly solutions do not necessarily prevent later disassembly. Both the manufacturer and the municipal maintenance actor indicated that pop rivets can be removed relatively easily in practice, for example by drilling them out. However, the technical manager also stressed that changing the joining method would require new testing, since the product would no longer be considered the same certified door. As he explained, if the joining method is changed, *"it is not the same product."* This indicates that current design solutions remain in place not only because they function technically, but also because existing approval and certification requirements support them. As a result, the present construction is not only shaped by technical choices, but also by the formal requirements surrounding the product.

Overall, the findings in this section show that current steel door systems are shaped by a combination of regulatory requirements, standardized design logic, and production-adapted assembly solutions. The empirical material also shows that redesign and reuse are linked to these requirements and production-related conditions.

4.3 Material and maintenance conditions

The empirical material further showed that current material and maintenance conditions both enable and limit circularity in steel door systems. As shown in Figure 2, this section includes current material choices, end-of-life limitations related to material composition, and the practical conditions for repair, reuse, and recycling. Across the interviews, respondents described how materials are selected to meet fire safety, technical performance, manufacturability, and commercial requirements, while at the same time creating difficulties for disassembly, safe handling, reuse, and recycling at the end of the product's life.

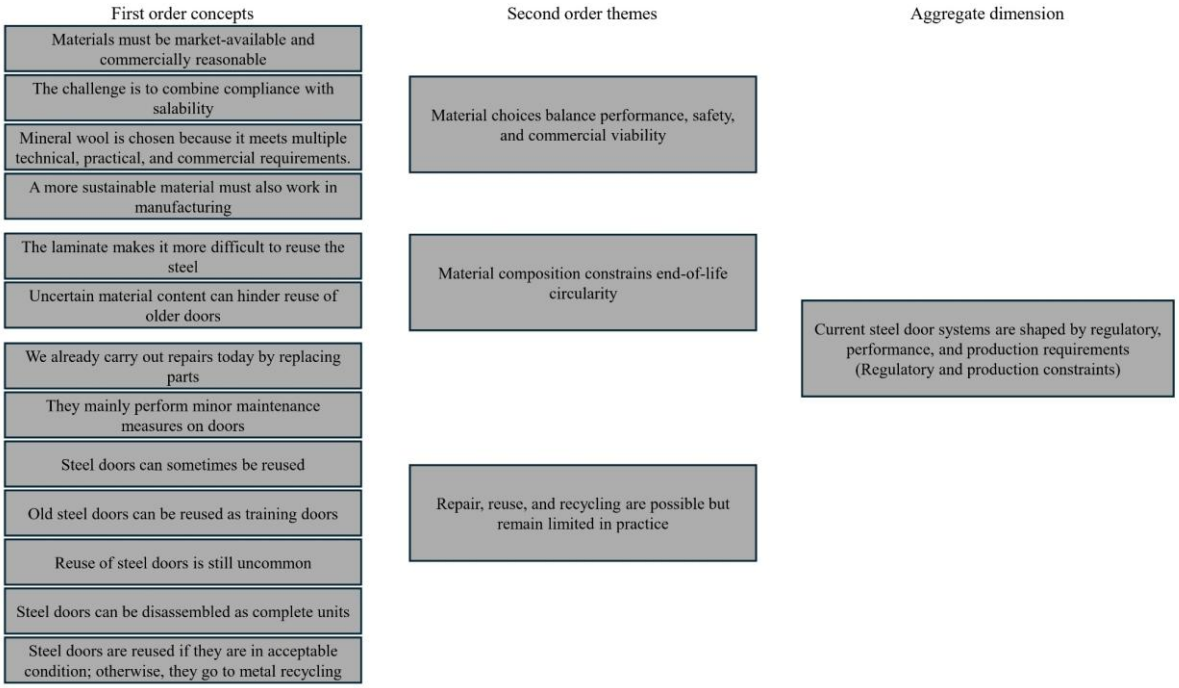


Figure 2: Material and maintenance conditions in the Gioia-inspired data structure

The analysis shows that current material choices are influenced by several requirements. According to the technical manager at the steel door manufacturer, material selection must meet fire related requirements as well as practical and commercial demands such as availability, handling, and cost. He explained that the materials used must be available on the market and reach a cost level that makes the product possible to sell and buy. For example, the choice of mineral wool reflects the need for a material that meets several demands at the same time, including fire performance, stability, sound insulation, and efficient production. As the same respondent explained, mineral wool is used because *"it is non-combustible, stable, and has a reasonable price."* The findings also show that more sustainable alternatives must be evaluated on more than environmental performance alone. Materials must work in the production process and provide the same required performance as current materials. Material selection therefore depends not only on technical suitability, but also on manufacturability and commercial viability.

The material-related findings also show that current material compositions can make circular handling more difficult at end of life. In particular, respondents described some surface treatments and material combinations as problematic because they make separation and reuse more difficult. The steel door manufacturer pointed to laminate as an example of a feature that may complicate the recovery of steel, since it is difficult to separate from the metal. In the municipal perspective, uncertainty about material content was also described as a practical barrier, especially in relation to older doors where hazardous substances may be present or where material history is unclear. As one interviewee explained, *"information is important"* in order to understand what the product contains and its history. This means that material composition affects circularity not only during manufacturing and use, but also in relation to handling, assessment, and recovery when doors are removed from buildings.

The data also showed that repair, reuse, and recycling are possible in some situations, but that these practices remain limited in scope and strongly dependent on context. Minor maintenance actions, such as cleaning, testing, adjusting frames, repainting, and replacing fittings, were

described as part of current practice. One interviewee described current interventions as limited, noting that *"these are not major interventions on a door,"* indicating that current interventions are mostly minor. Some respondents also pointed to cases where steel doors can be reused, for example in secondary applications or when the removed door remains in sufficiently good condition. At the same time, reuse was generally described as limited rather than common, and strongly dependent on factors such as the condition of the door, the required classification, and the practical feasibility of handling and relocation. In many cases, the outcome is therefore not direct reuse, but instead material recycling, especially when the door is no longer suitable for further use in its existing form.

Taken together, the findings in this section show that current material and maintenance conditions create both opportunities and barriers for circularity. On the one hand, some materials and maintenance practices already support durability, limited repair, and occasional reuse. On the other hand, these same conditions can also make disassembly, safe handling, reuse, and recycling more difficult in practice.

4.4 System-level conditions

The empirical material also showed that practical circularity in steel door systems depends on broader organizational and system-level conditions. As shown in Figure 3, this section includes the wider functional chain surrounding the door, the need for coordination, the role of information availability, and that practical implementation depends on project planning, time and available resources. Across the interviews, respondents described circularity not only as a question of product design, but as something influenced by the broader system.

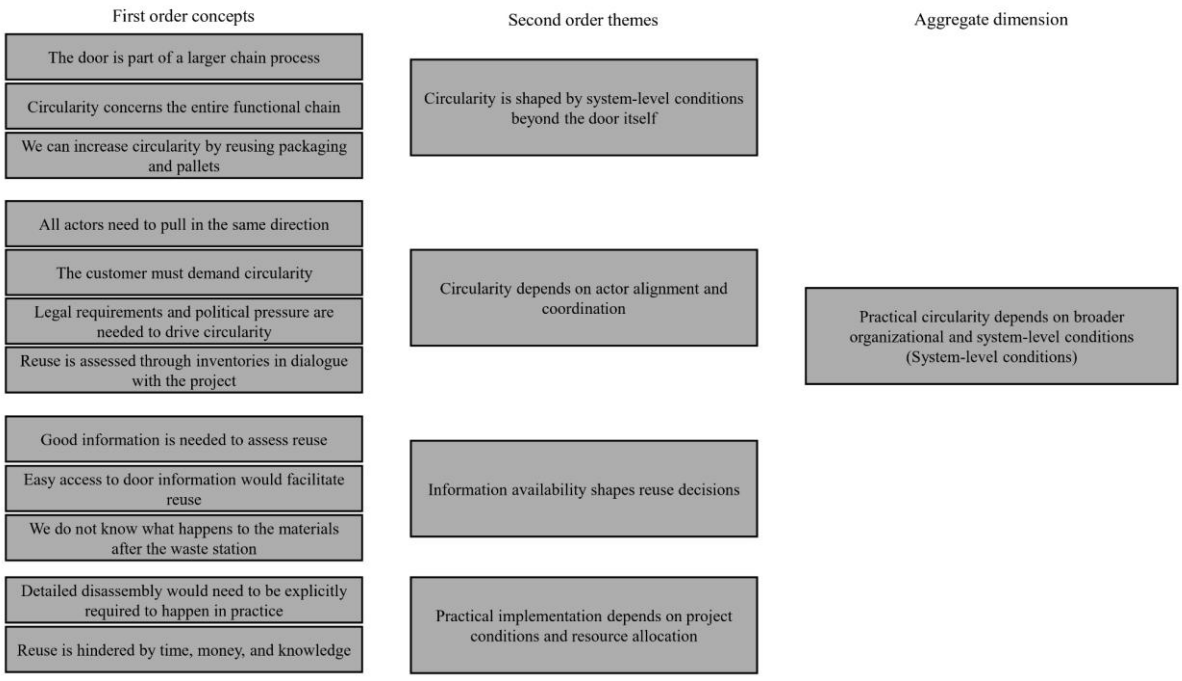


Figure 3: System-level conditions in the Gioia-inspired data structure

A recurring finding was that circularity must be understood in relation to the whole chain, rather than only the door itself. Respondents described the door as one element within a larger process that includes procurement, transport, installation, removal, and end-of-life handling. One respondent expressed this clearly by stating that *"the door is part of a chain process," while* another emphasized that circularity concerns *"the entire functional chain."* This broader

view was visible both in the manufacturer's and in the property management and reuse-related interviews. This also included logistics-related aspects, such as packaging and pallet reuse, which were described as additional opportunities for improving circularity beyond the product itself.

The empirical material further showed that circularity depends on coordination between multiple actors. Several respondents emphasized that circular solutions require alignment between customers, manufacturers, project organizations, and public actors. Customer demand was described as important, since circularity is unlikely to become a priority unless it is explicitly requested. At the same time, political pressure and legal requirements were also described as important drivers. As one respondent put it, *"there must be agreement between all actors in the process,"* while another stressed that *"there must be a requirement from the customer."* From a project and reuse perspective, circularity was also linked to the interaction between inventories, project planning, and design decisions, indicating that reuse is not handled by one actor alone, but through coordination across several roles.

Another strong finding concerns the importance of information. Respondents repeatedly highlighted the need for reliable information about product history, guarantees, material content, and performance classifications in order to assess whether a door can be reused. This was particularly clear in relation to older doors, where uncertainty about age, hazardous content, or technical properties makes reuse more difficult. One interviewee explained that *"information is important"* in order to understand what the product is, its history, and whether it may contain hazardous substances. In this context, product information in easily accessible forms, such as QR codes or similar documentation systems, was described as highly valuable.

The findings also show that practical implementation depends heavily on project conditions and resource allocation. Detailed disassembly, sorting, and recovery were not described as default practices, but as actions that would require explicit planning, procurement requirements, and sufficient time and budget. Several respondents pointed to cost, labor, competence, and time pressure as major reasons why circular practices are difficult to carry out consistently. One respondent explained that on construction sites *"they do not have time,"* in larger projects especially, speed and cost-efficiency were described as strong competing priorities, which means that potentially circular actions may be excluded unless they are clearly required and resourced from the start.

Overall, the findings in this section show that circularity in steel door systems is strongly conditioned by system-level factors beyond the door itself. Even when technical possibilities for reuse, disassembly, or recovery exist, practical implementation depends on information, coordination, procurement logic, and the practical conditions of the project environment.

4.5 Future circular design requirements

The empirical material finally pointed to a set of future-oriented design requirements for improving circularity in steel door systems. As shown in Figure 4, this section concerns the need for clearer and more accessible product information, as well as easier disassembly and handling at end of life. Across the interviews, respondents not only described current barriers, but also highlighted what would need to change for steel doors to become more practically reusable, traceable, and manageable in future projects.

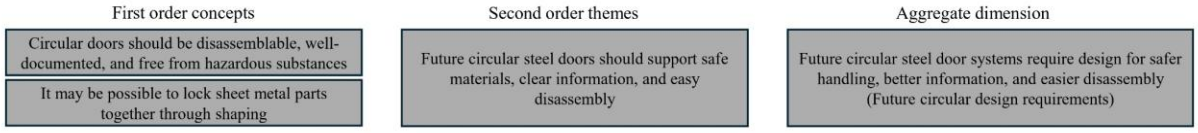


Figure 4: Future circular design requirements in the Gioia-inspired data structure

These findings also underscore the importance of clear product information. Across the interviews, respondents repeatedly stressed the need for reliable information on classification, material composition, performance, and previous use. This was described as especially important when assessing whether a removed door can be reused in another context. Easy access to this type of information, for example through QR codes, product passports, or similar documentation systems, was seen as highly valuable. One respondent described this possibility as *"very good"* because it would allow actors to *"get direct facts."* The findings therefore suggest that future circular steel doors should not only perform well in use but also include accessible and reliable product information that supports decisions on reuse, relocation, and end-of-life handling.

The empirical material also points to easier disassembly as an important future requirement. Although current joining methods were not always described as major barriers, several respondents still emphasized that disassembly must be fast, safe, and practical if circularity is to work in larger renovation and construction settings. This was connected to the broader finding that circular practices are unlikely to be prioritized if they require too much time, labor, or cost. One respondent noted that pop rivets can be removed *"very easily"* with *"a metal drill bit,"* suggesting that the main issue is not only technical removability, but whether handling and disassembly can be carried out under realistic project conditions. The findings therefore suggest that disassemblability is not only a technical feature, but also a practical condition for whether circularity can be implemented in real projects.

The findings in this section point to the importance of technical performance, clear documentation, and practical disassembly conditions in future circular steel door systems. Respondents highlighted not only fire safety and durability, but also the need for better identification, handling, recovery, and reuse in practice.

4.6 Stakeholder Mapping

Stakeholder mapping was used to visualize the actors involved in the steel door system and to clarify how circularity depends on more than the product design itself. The mapping includes actors connected to manufacturing, material and component supply, certification, sales and installation, property ownership, reuse platforms, recycling and end use. This includes CH Industry's suppliers, certification bodies, Svenska Skydd, KFast and property companies, reuse platforms such as CCBuild, recyclers and end users. To support this mapping, Table 5 presents the stakeholders involved in steel door products, together with their stakeholder category, role, influence on circularity, and relevance to circularity.

Table 5: Table of stakeholders involved in steel door products

Stakeholder	Role	Influence on circularity	Circularity relevance
CH Industry	Manufacturer and product owner	Defines design, materials, production and documentation	Central actor for circular design
Suppliers	Provide materials and components	Affect material content, replaceability and recyclability	Important for material choices
Certification bodies	Test and approve performance	Define what changes are allowed	Critical for certified reuse and redesign
Svenska Skydd	Sales and installation actor	Handles installation, replacement and removed doors	Important for reuse and take-back
KFast, property companies	Builds, renovates, owns and manages properties	Influences procurement, planning, reuse, storage and replacement	Key actor for circular project implementation
Reuse platforms, e.g. CCBuild	Match reusable products with new users	Support inventory and redistribution	Useful for reuse if data is reliable
Recyclers	Recover material from discarded doors	Determine recycling quality	Important for end-of-life recovery
End users	Use the doors during service life	Affect wear, damage and maintenance needs	Indirect role in product lifetime

Figure 5 gives an overview of the stakeholders and shows how they are positioned within the wider product system. CH Industry is at the center of the figure because it is the focal company in this study and acts as both manufacturer and product owner. The surrounding actors are divided into upstream actors, use phase/project-side actors, and end-of-life/recovery actors.

The purpose of the mapping is to show that circular strategies for steel safety doors involve coordination between several actors across the product life cycle, not only changes within CH Industry's product design. Suppliers and certification bodies influence material choices, components, testing and approval conditions. Property companies, Svenska Skydd and end users influence how the doors are installed, used, maintained, replaced and removed. Reuse platforms and recyclers become relevant when doors or components reach end of use and need to be assessed for reuse, redistribution or material recovery.

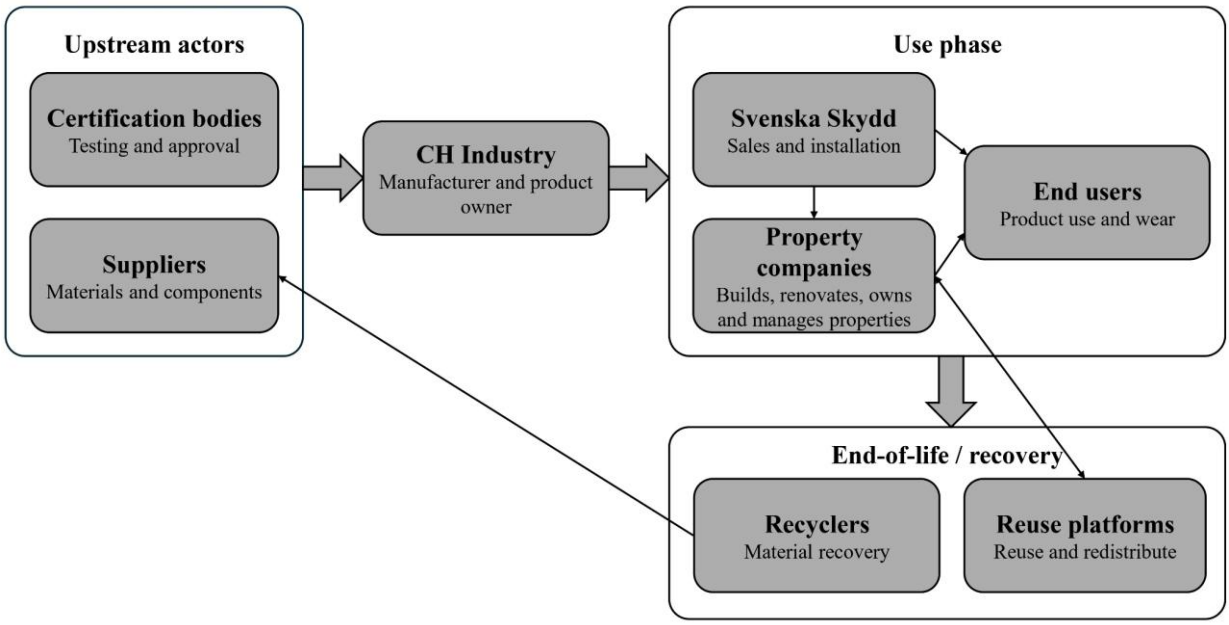


Figure 5: Overview of stakeholders involved in steel door products

4.7 Summary framework of circularity possibilities and design recommendations

The framework is based on the research questions and the analysis of empirical data. The current requirements and practices on the left side of the framework were identified through the Gioia-inspired analysis. The framework then shows how these requirements and practices affect circularity and how the findings are translated into circular design recommendations.

The left side addresses the first part of RQ1 by identifying the main requirements and practices that influence steel door design. These include certification and performance requirements, production practices, product architecture, material choices, maintenance conditions, documentation, and system-level factors in the value chain.

The middle part addresses the second part of RQ1 by analyzing how these factors support or limit circularity. Supporting factors include long service life, durable materials, maintenance practices, standardized components and existing product documentation. Limiting factors include strict certification boundaries, limited traceability, permanent joining methods, missing product history, and uncertainty about whether removed doors or components can be reused.

The right-side addresses RQ2 by translating the findings into circular design recommendations. These focus on maintaining verified performance, improving maintenance and service conditions, clarifying product architecture and intervention boundaries, strengthening product information and traceability, and improving disassembly and end-of-life handling.

Overall, the framework shows the connection between current design requirements, their effects on circularity, and the design recommendations developed in this study. It does not present a complete technical redesign. Instead, it shows which requirements and practices affect circularity and where future design improvements can be made while respecting certification, performance and production constraints.

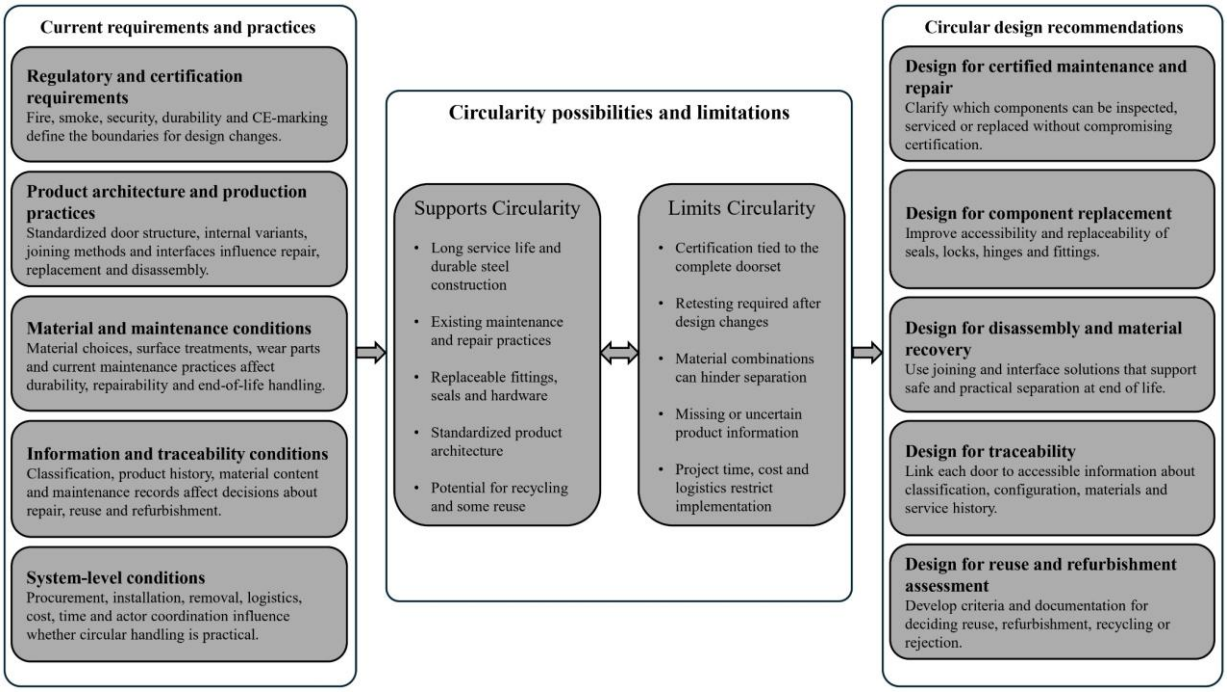


Figure 6: Framework linking current requirements and practices, circularity possibilities and circular design recommendations for steel doors

5. Analysis and Discussion

This chapter presents an analysis and discussion of the study's findings. The chapter is organized around the main themes identified in the study and follows a consistent structure: findings, theory, benchmarking, discussion and contribution.

5.1 Design requirements, Industry Practices, and Circularity in Steel Doors

To answer RQ1: "What current design requirements and practices influence steel door design, and how do they support or limit circularity?", the following sections discuss the main factors that influence the possibilities for circular design in steel safety doors. These factors are grouped into certification and performance requirements, production practices and product architecture, material and maintenance conditions, and broader system level factors.

5.1.1 Certification and performance requirements

The empirical findings indicate that certification and performance requirements are the strongest design boundaries for steel safety doors. Fire safety, smoke tightness, burglary resistance, sound insulation, and durability requirements influence both current product design and the possibilities for future redesign. This means that circularity cannot be introduced independently from approval, testing, and classification requirements.

Even when a door remains functional in its existing building, reusing it in another building may be difficult. The door may have been approved according to the requirements that applied during its original certification and may not meet the requirements that apply for the new use context. Previous interventions, such as maintenance, component replacement, or other modifications, may be difficult to verify. A barrier to reuse is therefore not only the door's functionality but also its classification, approved configuration and continued performance.

This creates challenges for direct application of circular strategies from 10R hierarchies. Strategies such as repair, refurbishment and remanufacturing retain more product value than recycling (Reike et al., 2018). However, the findings indicate that certification does not constrain these strategies equally. Repair focuses on restoring the product's original function by repairing or replacing damaged parts. Refurbishment and remanufacturing involve broader changes, such as disassembly, and component repair or replacement (Reike et al., 2018; Paul et al., 2024). For steel safety doors, such changes would require testing to verify the continued performance of the door. Each door needs to be assessed as a complete system, since the interaction between the door, its components and the surrounding wall can affect fire performance (Prieler et al., 2023). Therefore, the feasibility of each strategy must be assessed based on the door's condition, the extent of the changes made, and whether its approved configuration and continued performance can be verified.

The benchmarking supports this pattern. Among the reviewed cases, Daloc presented one of the broadest circular offers, including limited reuse practices, but its reused doors were offered without classification (Daloc AB, 2025). This suggests that reuse may be technically possible while still being restricted by certification and performance requirements. The ReDoor project further reinforces this point, as it explores whether the reuse of doors can be combined with verified fire safety performance (DBI, 2025). Together, these benchmarked examples support the empirical finding that certification is not only a design requirement, but also a key

boundary for whether higher-value circular strategies, such as reuse, can be implemented in practice.

Certification and performance requirements therefore both support and limit circularity. They support circularity by ensuring durability, safety, and long service life. At the same time, they limit circularity when continued performance cannot be verified following repair, reuse, refurbishment, or remanufacturing. This indicates that circular strategies in steel safety door manufacturing must account for both certification and performance requirements before they can be implemented in practice.

The contribution of this thesis is therefore to show that circularity in steel safety doors must be understood as a design challenge shaped by certification and performance requirements. The findings indicate that circularity is not only a question of materials or end-of-life handling. Circular strategies must also align with product approval, documentation, and performance verification. This contributes to a more context-specific understanding of circular design in regulated steel door manufacturing and helps explain why higher-value strategies such as reuse depend not only on technical feasibility, but also on verified compliance.

5.1.2 Production practices and product architecture

The current steel safety doors are based on relatively standardized architecture. Different classifications can be achieved using the same outer “shell”, while internal components such as insulation and reinforcements are adjusted to meet specific requirements. This indicates that the current design already contains a form of product platform logic, in which a base structure is adapted to different classifications and customer needs.

This can be related to product architecture, where product functions are allocated to physical components and connected through interfaces (Ulrich, 1995). In steel safety doors, performance requirements are not only linked to individual components, but also to how the door system functions as a complete assembly. The door leaf, frame, insulation, reinforcements, hinges, locks, seals and surface materials can all influence whether the doorset meets its required classification. Circular design must therefore address the product as a whole, because changes to individual parts may affect the certified performance of the complete assembly.

The findings also relate to modularity theory. Modularity depends on standardized and decoupled interfaces, which can support maintenance, component replacement, and upgrading (Fixson & Park, 2008). However, the findings indicate that existing standardization mainly supports different door configurations and does not necessarily make components easier to access, replace, or separate. The doors were primarily designed for safety, certification, production, and long service life, rather than future remanufacturing. Given the long service life of steel safety doors, future systems for reuse and remanufacturing, component availability, and certification requirements are uncertain. Future steel safety-door designs should therefore support later maintenance, component replacement, and disassembly without relying on a specific reuse or remanufacture. This means that circular design should support interventions without relying on predetermined recovery processes. Circular design guidelines recommend dividing products into modules or subcomponents to support assembly, disassembly, repair, and failure identification (Mesa, 2023). A modular platform with standardized components and interfaces can also simplify disassembly and reassembly (Paul et al., 2024). This suggests that future steel-safety door designs do not need to begin with a completely new door construction.

A more realistic direction is to build further on the existing standardized architecture by identifying which parts of the door should remain as the base structure and which could be made more replaceable, serviceable, or easier to separate.

The benchmarking supports this interpretation. Novoferm presents standardized fittings and retrofittable components as part of its circular approach, which suggests that modularity-related principles can support maintenance, component replacement, and upgrading in steel door systems (Novoferm Group, n.d.). However, the example concerns selected components rather than fully modular door architecture. This example suggests that existing door design can be adapted to support circular activities during the use phase without requiring a completely new door construction.

At the same time, the current architecture also limits circularity because certified performance depends on the complete product configuration. Circular design should therefore focus on components that can be replaced, repaired, or removed without compromising the approved configuration and performance of the complete doorset.

Overall, this finding indicates that circularity in steel safety doors should not be understood only as a matter of material choice or end-of-life treatment. It should also be understood as a product architecture challenge. The findings suggest that the existing standardized architecture can provide a foundation for circular design, but only if modularity is developed selectively within the limits set by certification and performance requirements.

5.1.3 Material and maintenance conditions

Material choices are mainly influenced by fire performance, manufacturability, cost, availability, and durability. The empirical findings suggest that sustainable material alternatives must be evaluated on more than environmental performance alone. They must also meet technical and industrial requirements. Mineral wool, for example, was identified as a material that meets several technical, functional, and production-related requirements at the same time.

This finding can be related to circular design strategies. Bocken et al. (2016) argue that circular design strategies need to function within existing technical and organizational constraints. In steel safety doors, long-life materials and durable constructions can slow resource loops by keeping the product in use for longer, while maintenance and repair can further extend the product's service life (Franconi et al., 2022). Material combination, surface treatments, and laminates may hinder recycling when they cannot be easily separated at end of life. This makes separable connection and suitable material choice important for disassembly, reuse, remanufacturing, and recycling (Mesa, 2023; Formentini et al., 2025). From a 10R perspective, repair and reuse are preferable because they retain more of the product's original value than strategies such as recycling (Reike et al., 2018).

The benchmarking indicates a similar pattern across the sector. Circularity in the reviewed companies was expressed mainly through steel recyclability, environmental documentation, and long service life, rather than through more developed systems for reuse or refurbishment. Because steel is the dominant material in these products, companies can present a high degree of circularity by emphasizing recyclability. Teckentrup and ASSA ABLOY, for example, mainly address circularity through environmental documentation, which provides information on materials, environmental performance over the product life cycle, and end-of-life scenarios (ASSA ABLOY Security Solutions Metal Industries Factory [LLC], 2019; Teckentrup GmbH

& Co.KG, 2012). The documentation mainly supports environmental transparency and provides limited information on how doors or components could be reused or refurbished, which is expected since such documentation primarily reports environmental performance.

Taken together, the empirical findings and benchmarking indicate that current material choices support circularity mainly through durability, long service life, recyclability, and some level of maintenance. These approaches support circularity in different ways. Durability and maintenance slow resource loops by extending the period during which the door remains in use, whereas recyclability supports material recovery after the door reaches the end of its service life (Bocken et al., 2016; Franconi et al., 2022). Current practices combine product life extension with recycling at end of life. The aim, however, should be to prioritize product life extension where feasible, since it preserves more of the product's original function and value (Reike et al., 2018). The findings also show that it's possible to replace fittings, seals, and locks in order to extend the service life of the door. However, maintenance and repair are mainly treated as service activities rather than as part of a wider approach to reuse or refurbishment.

The contribution of this analysis is that material circularity in steel safety doors should not be understood only as a question of recyclability or environmental performance. It must also be understood in relation to technical requirements, maintenance potential, and certification-related constraints. Material choices therefore influence not only environmental performance but also the possibilities for maintenance, reuse, and end-of-life handling in regulated steel door systems.

5.1.4 System-level conditions and stakeholder coordination

The empirical findings indicate that circularity depends on more than the physical design of the door. Steel safety doors are part of a wider system involving manufacturers, suppliers, installers, property owners, contractors, customers, certification bodies, and end of life actors. Circular outcomes therefore depend on coordination across this whole system.

This finding supports the circular economy perspective described by Geissdoerfer et al. (2017). From this perspective, circularity is not only a product issue but also a matter of production, use, and recovery systems. Information sharing is also important. Circular decisions depend on information about the product, its maintenance, materials, return process, and compliance with relevant requirements (Jensen et al., 2023). For steel safety doors, even a technically reusable door may not be suitable for reuse. This can happen when there is no actor responsible for assessing it, storage or documentation is unavailable, customer demand is lacking, or the project has insufficient time. This is also consistent with the literature, which indicates that missing information about condition, maintenance, spare parts, and reuse potential may make it difficult to determine whether a product is suitable for reuse (Çetin et al., 2023). The findings therefore suggest that technical reuse potential does not guarantee reuse in practice.

The benchmarking supports this perspective. The reviewed companies mainly communicated circularity through recyclability, environmental transparency, and a few limited examples of reuse or adaptability, while broader circular business models were less common. This suggests that circularity in the steel door sector is still at an early stage, since current practices mainly focus on recyclability and environmental documentation, while reuse, refurbishment, and broader circular business models remain less developed. Together with the interview findings, this suggests that material recovery is more established than approaches that keep doors and components in use. Slowing and closing resource loops provide a useful perspective on this.

Slowing loops keep doors and components in use, whereas closing loops recover materials through recycling (Bocken et al., 2016). This means that circularity in the sector is currently focused on material recovery rather than on retaining the product's function and value (Reike et al., 2018).

Taken together, these findings indicate that system-level factors both support and limit circularity. The interview data highlighted the need for reliable information about product history, classification, material content, and previous use. Without such information, informed decisions on reuse, repair, refurbishment, and end-of-life handling become more difficult. This could potentially be addressed through traceability, meaning that the product and its components could be connected to digital information that supports decisions throughout the product's life cycle (Jensen et al., 2023; Markou et al., 2025). Circular practices can be supported through traceability systems, but their implementation still depends on sufficient resources and coordination between actors.

The contribution of this finding is that circularity in steel safety doors should not be understood only as a question of product design or material recovery. It must also be understood as a system-level issue shaped by conditions beyond the product itself.

5.2 Design Recommendations for Circular steel safety doors

To answer RQ2: "How can circular strategies be translated into design recommendations for future steel safety doors, including end-of-life considerations, given the identified requirements and constraints?", this section discusses how the identified requirements and constraints can be translated into design recommendations for future steel safety doors. The recommendations build on the empirical findings and the preceding analysis and focus on how circular strategies can be adapted to a product context where certification and performance remain central. Rather than presenting idealized circular solutions in isolation, the section highlights design directions that appear both relevant and realistic in relation to the studied case. Particular attention is given to how future door systems can better support maintenance, traceability, disassembly, and end-of-life handling while still meeting required classifications and industrial conditions.

5.2.1 Verified performance as a starting point for circular design

Building on the certification and performance analysis in Section 5.1.1, this recommendation focuses on how circular interventions can be developed without compromising verified performance. Since fire safety, smoke tightness, burglary resistance, sound insulation, and durability define the approved doorset, future circular design needs to work within these performance boundaries from the beginning.

This issue can be linked to Design for X, which considers specific design goals and constraints early in product development (Kuo et al., 2001). Design for maintenance and design for disassembly are especially relevant in this context because they support product life extension and planned interventions later in the product's life cycle (Sassanelli et al., 2020; Formentini et al., 2025). In the studied case, these approaches must be applied within the certification and performance boundaries of the complete doorset. This means that possible maintenance and disassembly interventions should be identified and considered during the design process.

The benchmarking provides practical support for this recommendation. For example, Novoferm's use of retrofittable components suggests that selected adaptations can be

introduced within existing steel door systems, and the ReDoor project indicates that there is an interest in the sector in investigating whether door reuse can be combined with verified fire performance (Novoferm Group, n.d.; DBI, 2025). Together, these examples suggest that circular development is being explored, but implementation must remain compatible with certification and verified performance.

Based on these findings, future steel safety doors should support circular intervention within clearly defined performance boundaries. During the product development process, intervention boundaries should be defined to clarify which interventions can be made within the approved configuration and which would require additional assessment or testing.

This would provide a clearer basis for developing circular solutions without compromising the verified performance of the complete doorset and suggest that verified performance provides a clear starting point for deciding which circular interventions are suitable in future steel safety door development.

5.2.2 Product architecture and intervention boundaries

Following the product architecture analysis in Section 5.1.2, the existing standardized door structure provides a possible basis for circular design. However, this potential depends on clearly defined components and interfaces. Future steel safety doors should therefore make it clearer which parts can be accessed, serviced, replaced, or separated without affecting the certified performance of the doorset.

This can be linked to product architecture. In the studied case, the distinction between the outer structure and the internal performance-related components suggests some potential for modularity. Standardized and decoupled interfaces can support maintenance, component replacement, and upgrading over time (Fixson & Park, 2008). Modular product architecture with standardized components and interfaces can also simplify disassembly and support product longevity and upgrading (Paul et al., 2024). When applied to steel safety doors, clearly defined and separable interfaces could allow selected parts to be repaired, replaced, or upgraded without replacing the complete door, provided that the certified configuration is not compromised.

Examples from the benchmarking point in a similar direction. Novoferm's retrofittable components and standardized fittings show how selected components can be adapted within existing steel door systems (Novoferm Group, n.d.). Daloc's design for disassembly and maintenance-related practices point in a similar direction (Daloc AB, 2025). This suggests that future steel safety doors can build on existing door design by making selected interfaces and serviceable components more accessible and easier to replace without changing the overall concept of the certified door.

Based on this, future steel safety doors should be developed with greater attention to interface clarity and clear boundaries for intervention. This means that the different parts of the door should not only support its function, but also be designed so they can be accessed, assessed, and, where relevant, replaced without unnecessarily affecting the rest of the doorset. In this way, product architecture becomes an important design condition for enabling circular interventions within the limits of certification and performance requirements.

This adds to the discussion by showing that circular design in steel safety doors can build on the existing product architecture. It suggests that product architecture has the potential to support circular interventions while maintaining verified performance.

5.2.3 Maintenance, service, and life extension during use

Based on the maintenance practices identified in Section 5.1.3, future steel safety doors should make service interventions such as the replacement of fittings, seals, and locks easier to plan, approve, and document. This would shift maintenance from being mainly a practical response during use to becoming a more deliberate circular design strategy. By supporting access to serviceable components and clarifying which replacements are allowed, future door designs can help extend the product's functional life while maintaining certification and performance requirements.

This can be related to circular design approaches that support maintenance and repair. These kinds of interventions contribute to product life extension (Franconi et al., 2022). From a Design for X perspective, design for maintenance supports repair and component replacement, which helps extend the product's use phase (Sassanelli et al., 2020). This suggests that maintenance should be integrated into product development rather than treated only as a response to wear and tear. For steel safety doors, existing maintenance and replacement practices offer a clear basis for further circular development.

In the benchmarking examples, circularity was communicated mainly through recyclability, environmental documentation, and long service life, while more developed systems for repair or refurbishment were less visible. However, Novoferm's use of retrofittable components and Daloc's emphasis on maintenance indicate that support for interventions during the use-phase already exists in the sector (Novoferm Group, n.d.; Daloc AB, 2025). This suggests that strengthening maintenance is a realistic direction in comparison to broader systems for reuse or refurbishment.

From this perspective, future steel safety doors should be developed in ways to make routine maintenance easier to perform. This does not require a completely new product concept but does require clearer support for maintenance throughout the product's lifetime. Future designs should therefore provide clear access to serviceable parts and guidance for maintenance and repair. This would make it easier to keep the doors in use for longer and help avoid unnecessary replacement.

This indicates that maintenance and service during use are important for improving the circularity of steel safety doors. This is particularly relevant because maintenance can build on existing practices and is therefore more feasible within the current product system than developing more advanced systems for reuse or refurbishment.

5.2.4 Product information, traceability, and material transparency

Based on the system-level traceability issues discussed in Section 5.1.4, product information, traceability, and material transparency are important conditions for supporting circular activities throughout the product's life cycle. The empirical material indicates that decisions about reuse, maintenance, and assessment depend on knowing what the door is, how it is classified, what it contains, and how it has been used. It also indicates that uncertainty about

material content and end-of-life handling makes assessment and recovery more difficult. This suggests that reliable and accessible information is a key condition for enabling circular activities throughout the product life cycle.

This can be linked to traceability, where a physical product or component is connected to digital information that supports decisions throughout the product life cycle (Markou et al., 2025). Product passports can also provide information relevant to maintenance, material content, and end-of-life handling (Jensen et al., 2023). For steel safety doors, this means that traceability can reduce uncertainty when deciding whether maintenance, reuse, or recovery is suitable, since these decisions depend on reliable information about the specific door and its history.

In the benchmarking examples, environmental product declarations are already used to communicate material and life-cycle information. However, these documents mainly support environmental transparency at the product level, while maintenance, reuse assessment, and recovery require more detailed information. Future traceability systems should therefore connect the physical door to data about classification, approved component combinations, maintenance history, and previous component replacements. This would make the information more useful for circular decisions such as repair, reuse assessment, refurbishment, and end-of-life handling.

Based on these findings, future steel safety doors should be designed with greater attention to traceability, documentation, and material transparency. Future door systems would benefit from accessible and durable product information on classification, material content, maintenance history, and possible limits for intervention. Since QR codes are already used on the doors, this could be further developed to provide more structured product information throughout the product life cycle. This would reduce uncertainty when assessing maintenance, repair, reuse, and end-of-life handling and support more informed decisions about the continued use and recovery of the door and its components.

This indicates that reliable and useful product information is essential for circular decision-making throughout the product's life cycle. For steel safety doors, this information is needed to determine which circular options are feasible while maintaining knowledge of the door's classification, condition and history.

5.2.5 Disassembly and end-of-life handling

Based on the end-of-life and handling conditions discussed in Section 5.1.4, controlled disassembly is important for future steel safety doors. The empirical material indicates that current joining methods do not appear to be the main barrier themselves. Instead, circular outcomes depend on whether disassembly and handling are feasible under realistic project conditions. This suggests that future circular design should focus not only on whether the product can be taken apart in theory, but also on whether this can be done in ways that support practical assessment, sorting, recovery, and possible reuse.

This can be related to design for disassembly, which aims to facilitate separation so that components and parts can be reused, recycled, or recovered (Ostapska et al., 2024). The effectiveness of disassembly depends on design factors such as connector types, tool requirements, access to components, and the ability to separate components without damaging parts intended for reuse (Formentini et al., 2025). For steel safety doors, this means that

disassembly should be assessed not only by whether the door can technically be taken apart, but also by whether this can be done without unnecessary damage and under realistic project conditions. This suggests that design for disassembly should focus on parts where separation creates a realistic opportunity for reuse or recovery.

In the benchmarking examples, design for disassembly was less visible than recyclability and environmental documentation. However, among the reviewed companies, Daloc was the clearest example of a company communicating some level of design for disassembly and maintenance-related practices (Daloc AB, 2025). This suggests that design for disassembly is being considered but is not yet as established as material recovery approaches.

From this perspective, future steel safety doors should be designed to support controlled disassembly and end-of-life handling without causing unnecessary damage to the door or reusable parts. The issue is not only whether the door can be taken apart, but also whether end-of-life processes can be made more practical, predictable and compatible with time and resource constraints in real projects. Circular design is thus closely linked to how the door can be handled after use, not only to how it performs during use.

This recommendation indicates that circularity in steel safety doors depends on practical end-of-life conditions, not only on theoretical disassembly potential. It highlights that disassembly, recovery, and reuse require handling processes that work under realistic time, cost and project constraints.

5.3 Practical implications

Circularity in steel safety doors should not be treated mainly as an end-of-life recycling issue. The findings indicate that circularity is also connected to certification, product information, maintenance and end-of-life handling across the product life cycle.

For the steel door sector, this means that circular strategies need to be developed in relation to verified product performance. Repair, reuse, refurbishment and remanufacturing become difficult to apply when the classification and approved configuration of the doorset cannot be confirmed. Therefore, the sector may benefit from clearer routines for distinguishing between safety-critical components and components that can be maintained, replaced or recovered.

The findings also indicate that circularity depends on information transfer between actors. Manufacturers, installers, property owners, maintenance actors and end-of-life actors need access to reliable information about classification, material content, approved components, maintenance history and possible intervention limits. Without this information, doors with potential for continued use or recovery may still be difficult to assess.

This suggests that the steel door sector should first focus on the conditions that make circular decisions more practical: documentation, traceability, maintenance support and clearer intervention boundaries.

5.3.1 Implications for CH Industry

For CH Industry, the identified design recommendations differ in how immediately feasible they appear within the current product system. Some directions could be developed within the existing architecture and documentation practices, while others would require more extensive

product development, additional testing, or closer coordination with external actors. For this reason, the implementation directions are discussed here from more immediately feasible measures to more demanding longer-term developments.

The more immediately feasible implementation measures concern documentation, traceability, and clarification of intervention boundaries. Improving access to information on classification, material content, maintenance history, and possible intervention limits could strengthen both maintenance and future end-of-life assessment without requiring immediate changes to the certified product architecture. A closely related step would be to further develop the existing QR codes on the doors so that they provide access to more structured product information. Such a solution could make product information more accessible in practice and could also provide guidance for maintenance, disassembly, and end-of-life handling. Another feasible measure would be to clarify internally which components are directly tied to certified performance and which may be more open to replacement or servicing. Together, these measures appear realistic because they could improve intervention clarity and reduce uncertainty without immediately requiring major changes to the door's physical design.

A second group of implementation measures concerns maintenance and life extension within the current product system. The findings suggest that CH Industry could strengthen circularity by further developing existing maintenance-related practices, for example through clearer support for replacement parts, service-related guidance, and a more structured maintenance logic for fittings, locks, and seals. These measures appear relatively feasible because they build on interventions that already occur in practice, even if they are not yet fully structured as circular strategies. In this sense, circular development could begin by strengthening the door's continued usability before focusing on more advanced recovery strategies.

More demanding longer-term implementation directions concern product architecture and end-of-life handling. Developing a door system with more explicitly replaceable components, more advanced support for controlled disassembly, or clearer design-for-disassembly features would likely require more extensive product development and may also affect the certified product configuration. Such directions may therefore be less immediately feasible within the current system, but they remain relevant as longer-term development goals if CH Industry aims to strengthen circularity more substantially over time.

Overall, this suggests that the most realistic path for CH Industry is to start with information, traceability, and clear intervention boundaries, and then move step by step toward more advanced circular product solutions. In this sense, practical implementation depends less on one major redesign and more on a phased development process in which early feasible measures lay the foundation for longer term circular improvements.

6. Conclusions

This chapter presents the conclusions of the study in relation to the aim and research questions. It summarizes the main findings concerning current design requirements, circularity limitations, and future design recommendations for steel safety doors. The chapter also presents the study's limitations and suggestions for future research.

6.1 Main findings and conclusions

The aim of this thesis was to map the current design requirements for steel safety doors in buildings and to develop recommendations for improving circularity in future steel door systems. To address this aim, the study examined how current design requirements and practices influence steel door design and how circular strategies can be translated into design recommendations under the constraints identified in the case. The study was based on a qualitative single case study that included interviews with CH Industry as the steel door manufacturing company, as well as related actors in the value chain, together with a benchmarking of circularity practices in the steel door sector. This provided the basis for understanding both the current conditions for circularity and the design directions that may support future improvement.

To answer the aim of this thesis, the following two research questions were addressed:

RQ1: What current design requirements and practices influence steel door design, and how do they support or limit circularity?

RQ2: How can circular strategies be translated into design recommendations for future steel safety doors, including end-of-life considerations, given the identified requirements and constraints?

Regarding RQ1, the study shows that current steel safety door design is strongly shaped by certification, fire safety, performance requirements, and production logic. These factors define the main boundaries within which the product is developed and maintained. Circularity is therefore not something that can be introduced independently from the existing approval and classification structure. The findings also show that some current practices already support circularity to a limited extent, particularly through durability, maintenance, replacement of selected components, and the standardized architecture of the door system. At the same time, higher-value circular strategies such as reuse, refurbishment, and repair beyond routine maintenance remain constrained in practice, since circular outcomes depend not only on the physical condition of the door, but also on whether its classification, condition, and documentation can be verified in a new context. The study also shows that circularity is shaped by broader system level factors, including information availability, traceability, actor coordination, project conditions, and practical end-of-life handling.

Regarding RQ2, the study shows that circular strategies can be translated into design recommendations for future steel safety doors, but only if they are adapted to the product’s regulatory and industrial context. The findings suggest several important design directions for future circular steel safety doors, with a focus on verified performance, maintenance, traceability, clearer material information, and more controlled end-of-life handling. Rather than suggesting a completely new door concept, the findings indicate that circular development should build on the existing product logic while making the system easier to maintain, assess,

document, and manage over time. In this sense, future circular steel safety doors should not only meet required performance and certification requirements, but also be better prepared for maintenance, intervention, recovery, and informed end-of-life decisions.

For CH Industry, the study suggests that the most feasible starting points are measures that can be developed within the current product system without immediate redesign or recertification, including improved documentation, stronger traceability, a clearer distinction between performance-critical and replaceable components, and better support for maintenance-related interventions. More advanced circular strategies, such as expanded design for disassembly or more adaptable product architecture, appear more relevant as longer-term development directions.

Overall, the thesis concludes that improving circularity in steel safety doors requires more than emphasizing recyclability. Circularity in this product category depends on how design, certification, information, maintenance, and end-of-life handling are connected across the product's life cycle. The contribution of the study is therefore to show that circular development in steel safety doors must be understood as both a product design issue and a system coordination issue. Future progress depends not only on what is designed into the door itself, but also on how the surrounding structures for documentation, assessment, maintenance, and recovery are developed.

6.2 Limitations and future research

This thesis has contributed by mapping current requirements and practices that influence steel safety door design and by developing recommendations for how circularity can be improved in future steel door systems. The study has shown that circularity in this product context is closely connected to certification, product information, maintenance, traceability, actor coordination, and practical end-of-life handling. However, the results should be understood in relation to the limitations of this study.

The first limitation is that the study is based on a limited number of actors and interviewees. The empirical material mainly includes CH Industry and closely connected stakeholders including Svenska Skydd and KFast. These actors provided relevant perspectives on manufacturing, sales, installation, property ownership, maintenance and reuse, but they do not represent the whole steel door value chain. Future research could therefore include a broader range of actors, such as suppliers, certification bodies, contractors, recyclers, customers, and additional property owners in order to gain a more complete understanding of the conditions that influence circularity across the full life cycle of steel safety doors.

A second limitation is that the thesis is based on one company and one specific industrial case. This made it possible to develop an in-depth understanding of CH Industry's steel safety doors, but it also means that the findings are contextually grounded rather than statistically generalizable. The results may be especially relevant for similar small or medium sized manufacturers operating in certified steel door contexts. But further studies are needed to examine whether the findings also apply to larger companies or other steel door manufacturers.

A third limitation is that the study is mainly based on qualitative insights rather than detailed technical design work. The results are based on interviews, observations, notes, photographs, benchmarking, and literature. But the thesis does not include detailed engineering calculations, material testing, certification testing, or complete technical specification. This also means that

no complete technical specification for a redesigned door is provided. Future research could therefore include more technical analysis of circular steel door design, for example, by examining specific components, solutions, material alternatives, joining methods, disassembly procedures or documentation systems in greater detail.

Another limitation is that the thesis does not include physical implementation or prototype testing. The design recommendations should therefore be understood as analytically grounded proposals rather than technically verified solutions. Future research could involve action research, prototyping, or pilot testing in collaboration with a company in order to evaluate how the proposed circular design recommendations work in practice. This could include testing, traceability solutions, maintenance-oriented documentation, disassembly procedures, or component replacement strategies in real production or renovation contexts.

Certification and reuse also require deeper investigation. The study shows that certification is one of the main barriers for reuse, refurbishment and design changes in steel safety doors. However, the thesis does not fully examine how reused fire safety doors or replaced components could meet certification requirements in practice. Future research could therefore investigate which parts of a certified steel door system can be repaired, replaced or reused without compromising safety performance, and what type of documentation or testing would be needed to support verified reuse.

Finally, the thesis has only limited focus on policy and regulation. Circularity is discussed in relation to product design, documentation, certification, and practical implementation, but the study does not deeply analyze how regulations or policy instruments could be developed to support circular steel door reuse. Future research could therefore examine what regulatory support, procurement requirements or industry guidelines would be needed to make circular strategies such as reuse, refurbishment, traceability and take back more feasible in the steel door sector.

Overall, these limitations show that the thesis provides a strategic and conceptual basis for circular design and steel safety doors, but further research is needed to technically validate, test, and expand the recommendations. Future work should therefore focus on broader actor involvement, technical verification, certification related reuse conditions, and practical implementation in real projects.

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Appendices

The appendices contain supplementary material used in the thesis. Appendix A contains the interview guides used during the data collection. Appendix B presents the Gioia-inspired coding process. Appendix C contains photographs from the production walkthrough.

Appendix A - Interview guides (Swedish)

A1 Interview guide: CH industry / Technical manager

Intervjufrågor

1. Dörrarnas nuvarande konstruktion

Hur är dörrarna konstruerade idag på en övergripande nivå?

Följdfrågor:

- Vilka huvudkomponenter består en typisk ståldörr av?
- Vilka material används i de olika delarna av dörren och varför har just de materialen valts?

2. Material och komponenter

Finns det material eller komponenter som är särskilt svåra att byta ut ur ett tekniskt eller praktiskt perspektiv?

Följdfrågor:

- Är vissa materialval främst styrda av krav på brand, säkerhet, ljud eller hållbarhet?
- Finns det material eller komponenter som ni ser som problematiska ur ett cirkularitetsperspektiv?
- Använder ni idag återvunnet material i någon del av produkten, eller har det diskuterats?

3. Konstruktion och demontering

Vilka delar går relativt enkelt att ta loss eller byta ut?

Följdfrågor:

- Finns det komponenter som sitter permanent sammanfogade och därför är svåra att separera?
- Finns det någon del av dagens konstruktion som du tycker begränsar möjligheten till reparation eller återbruk?

4. Reparation och utbyte av komponenter

Vilka komponenter är vanligast att behöva reparera eller byta ut under dörrens livslängd?

Följdfrågor:

- Är dörrarna idag utformade för att sådana byten ska vara enkla att göra?
- Finns det skillnad mellan vad som är teoretiskt möjligt att byta och vad som i praktiken faktiskt görs?
- Vilka hinder finns det för att arbeta mer med reparation i stället för att byta hela dörren?
- Skulle det vara möjligt att utveckla dörrarna så att vissa komponenter lättare kan bytas ut utan att påverka dörrens övriga funktion?

5. Återbruk och cirkulära möjligheter

Hur ser du på möjligheten att återbruka hela dörrar i framtiden?

Följdfrågor:

- Vilka delar eller komponenter tror du har störst potential att återanvändas?
- Vad skulle krävas för att återbruk ska bli mer realistiskt i praktiken?
- Finns det några komponenter som skulle kunna renoveras eller återställas i stället för att kasseras?
- Om man skulle designa dörrar mer cirkulärt från början, vilka förändringar skulle vara mest relevanta?
- Ser du störst potential i reparation, komponentutbyte, renovering eller återbruk av hela dörrar?

6. Krav, begränsningar och certifiering

Vilka krav styr konstruktionen mest idag?

Följdfrågor:

- Hur påverkar brandklassning, säkerhetskrav och certifieringar möjligheten att förändra konstruktionen?
- Finns det förändringar som hade varit önskvärda ur ett cirkularitetsperspektiv men som blir svåra på grund av certifieringskrav?
- Hur ser du på balansen mellan säkerhet, prestanda och cirkulär design?

7. Spårbarhet och information

Hur dokumenteras idag information om dörrarnas material, komponenter och konstruktion?

Följdfrågor:

- Finns det idag tillräcklig information för att i efterhand avgöra hur en dörr är uppbyggd?
- Vad skulle behövas för att bättre kunna följa en dörr eller dess komponenter över tid?

- Tror du att bättre spårbarhet skulle kunna underlätta reparation, underhåll eller återbruk?

8. Avslutande frågor

Finns det något i dagens dörrdesign som du tycker fungerar särskilt bra ur ett livslängds- eller underhållsperspektiv?

Följdfråga:

- Finns det något du tycker borde utvecklas om man vill göra dörrarna mer cirkulära?

Frågeurval för intervju om konstruktion och materialval

- Kan du kort berätta om din roll och hur du arbetar med ståldörrarna idag?
- På vilket sätt är du involverad i dörrarnas konstruktion, materialval eller komponentval?
- Hur ser processen ut när en ny dörr eller dörrlösning tas fram?
- Hur är dörrarna konstruerade idag och vilka huvudkomponenter består de av?
- Vilka krav styr konstruktionen mest idag?
- Vilka typer av förändringar tror du skulle vara realistiska utan att kompromissa med kraven?
- Har ni tänkt på att dörren ska kunna demonteras eller inte?
- Hur ser ni på att göra fler delar åtkomliga för service eller utbyte?
- Varför valde ni just nitar och vad finns det för andra alternativ?
- Hur varierar konstruktionen mellan olika dörrtyper, exempelvis beroende på brandklassning eller säkerhetsklassning?
- Är komponenter standardiserade mellan modellerna?
- Vilka material används och varför?
- Är mineralull vald främst för brandsäkerhet, eller finns det även andra skäl bakom materialvalet?
- Vilka delar är lättast respektive svårast att reparera eller byta ut?
- Var ser du störst potential för cirkularitet: reparation, komponentutbyte, renovering eller återbruk av hela dörrar?
- Vad i dagens konstruktion begränsar möjligheten till reparation, komponentutbyte eller återbruk?
- Finns det något i dagens dörrdesign som du tycker fungerar särskilt bra ur ett livslängds- eller underhållsperspektiv?
- Finns det något du tycker borde utvecklas om man vill göra dörrarna mer cirkulära?

- Är det något vi inte har frågat om som du tycker är viktigt för att förstå dörrarnas konstruktion eller cirkulära potential?

A2 Interview guide: Svenska Skydd

Intervjufrågor med fokus på installation, demontering, service och återbruk

1. Installation av stålsäkerhetsdörrar

Kan du beskriva hur installationen av stålsäkerhetsdörrar vanligtvis går till i praktiken?

2. Demontering av gamla dörrar

Kan du beskriva hur demonteringen av gamla dörrar vanligtvis går till?

Följdfrågor:

- Uppstår det vanligtvis några skador på dörren i samband med demonteringen? Om ja, vilka?
- Finns det något i installations- eller demonteringsprocessen som påverkar möjligheten till cirkularitet?

3. Orsaker till att dörrar demonteras eller byts ut

Vilka är de vanligaste orsakerna till att en dörr demonteras eller byts ut?

Följdfrågor:

- Beror det på skador, renovering, slitage, brand, inbrott eller andra skäl?
- Varför tas gamla säkerhetsdörrar vanligtvis bort?
- Är de flesta borttagna dörrar tekniskt obrukbara, eller tas vissa bort trots att de fortfarande fungerar?

4. Dörrarnas skick vid demontering

Vilket skick brukar dörrarna vara i när de demonteras?

Följdfrågor:

- Vilka delar eller komponenter är det som oftast går sönder eller leder till att dörren måste bytas?
- Finns det vissa delar av dörren som oftare än andra är i tillräckligt gott skick för att kunna tas tillvara?

5. Hantering efter demontering

Vad händer normalt med dörrarna efter att de har demonterats?

Följdfrågor:

- Skrotas de, lagras de, skickas de vidare eller hanteras de på något annat sätt?

- Har du något konkret exempel på ett projekt där en dörr demonterades, och vad som hände med den efteråt?

6. Reparation, renovering och återbruk

När gamla dörrar demonteras, ser du några realistiska möjligheter att återanvända hela dörrar, renovera dem eller ta tillvara vissa delar eller material?

Följdfrågor:

- I vilka fall går en säkerhetsdörr att reparera i stället för att bytas ut helt?
- Vad är det som oftast hindrar reparation eller återbruk i praktiken?
- Vilka hinder ser du för återbruk, renovering eller återvinning av dörrarna eller deras komponenter?
- Vilka delar skulle vara mest realistiska att göra mer modulära eller enklare att byta?

7. Certifiering och krav

Hur påverkar certifiering möjligheten att byta delar, reparera eller återanvända en dörr?

8. Information och spårbarhet

Vilken information behöver ni för att kunna bedöma om en dörr kan repareras, återbrukas eller renoveras?

Följdfråga:

- Vilken information saknas oftast idag när ni möter äldre dörrar ute hos kund?

9. Framtida förbättringar

Om ni fick ändra tre saker i framtida ståldörrar för att underlätta service, reparation och återbruk, vad skulle ni välja?

A3 Interview guide: KFast

Huvudfrågor

1. Roller och perspektiv

Kan ni kort beskriva era roller och hur ni arbetar med cirkularitet, återbruk, förvaltning eller byggprojekt?

2. När dörrar byts ut

I vilka situationer brukar dörrar bytas ut i ert bestånd?

Följdfrågor:

- Är det oftast på grund av slitage, skador, säkerhetskrav, brandskydd, renovering eller estetiska skäl?
- Byts dörrar ofta trots att de fortfarande fungerar?

3. Beslut: behålla, reparera eller byta ut

Hur resonerar ni idag kring att behålla, reparera, uppgradera eller byta ut dörrar?

Följdfrågor:

- Vad väger tyngst i beslutet?
- Finns det någon skillnad mellan olika typer av byggnader eller projekt?

4. Vad händer med dörrar som tas bort?

Vad händer vanligtvis med dörrar som demonteras vid renovering eller ombyggnad?

Följdfrågor:

- Slängs de direkt, sorteras de, lagras de eller återbrukas de ibland?
- Finns det någon etablerad process för att bedöma om en dörr kan användas igen?

5. Erfarenhet av återbruk

Har ni erfarenhet av att återbruka dörrar eller andra byggprodukter i era projekt?

Följdfrågor:

- Har ni försökt återbruka dörrar specifikt?
- Vad fungerade bra respektive mindre bra?
- Finns det exempel där återbruk stoppades trots att det verkade möjligt?

6. Krav som påverkar möjligheten till återbruk

Vilka krav ser ni som mest avgörande när det gäller dörrar i byggnader?

Följdfrågor:

- Brandskydd?
- Inbrottsskydd?
- Ljud?
- Tillgänglighet?
- Standardisering och passform?
- Ansvar, garanti eller försäkringsfrågor?

7. Hinder för cirkulära dörrlösningar

Vilka är de största hindren för att återbruka eller på annat sätt arbeta mer cirkulärt med dörrar?

Följdfrågor:

- Tekniska hinder?
- Regelverk eller certifiering?
- Logistik och lagring?
- Tidsbrist i projekt?
- Osäkerhet kring skick eller dokumentation?
- Ansvarsfördelning mellan aktörer?
- Ekonomi eller upphandling?

8. Information och spårbarhet

Vilken information skulle ni behöva för att kunna bedöma om en dörr är lämplig att återbruka?

Följdfrågor:

- Brandklass?
- Säkerhetsklass?
- Ålder?
- Tillverkare?
- Tidigare placering?
- Servicehistorik?
- Skador eller modifieringar?
- Monteringsanvisningar?

9. Digital information och produktpass

Skulle bättre spårbarhet, till exempel via märkning, QR-kod eller digitalt produktpass, göra återbruk av dörrar mer realistiskt för er?

Följdfrågor:

- Vilken information skulle vara mest användbar?
- I vilket skede skulle den vara viktigast: inventering, demontering, projektering eller återinstallation?

10. Framtida cirkulära dörrlösningar

Om ni fick önska er förbättringar i framtida dörrsystem, vad skulle göra dem mer användbara ur ett cirkularitets- eller återbruksperspektiv?

Följdfrågor:

- Enklare demontering?
- Utbytbara komponenter?
- Standardiserade mått eller gränssnitt?
- Bättre dokumentation?
- Lättare bedömning av skick?
- Möjlighet till uppgradering i stället för fullständigt utbyte?

11. Aktörer och samverkan

Vilka aktörer behöver samarbeta för att återbruk av dörrar ska fungera bättre i praktiken?

Följdfrågor:

- Fastighetsägare?
- Tillverkare?
- Installatörer?
- Entreprenörer?
- Återbruksaktörer?
- Certifieringsorgan?
- Lagstiftare?

12. Avslutande rekommendation

Om ni skulle ge ett råd till en dörrtillverkare som vill utveckla mer cirkulära ståldörrar för fastighetssektorn, vad skulle det vara?

Appendix B - Gioia-inspired Coding

This appendix presents the Gioia-inspired coding process used in the analysis. The appendix is divided into three parts, showing how the interview data was coded from initial excerpts into first-order concepts, second-order themes, and possible aggregate dimensions.

Appendix B1: Initial Gioia Coding

This appendix shows the initial coding stage of the Gioia analysis. The table includes the original interview excerpts, cleaned meanings, tentative first-order concepts, memos, and merge notes.

Note on merge codes:
M1 = requirements, certification, testing, and approval
M2 = material choices, material limitations, mineral wool, and laminate
M3 = standardization and the same basic construction
M4 = joining method/joining
M5 = existing repair, disassembly, and circularity in practice
M6 = system perspective, value chain, packaging, actors, and policy/logics beyond the door itself

Due to the width of the initial coding table, it is presented in two parts. Part 1 includes the interview ID, respondent role, topic, and original excerpt. Part 2 includes the corresponding cleaned meaning, tentative first-order concept, memo, and merge note. The row number is included in both parts.

Part 1:

ROW	Interview_ID	Respondent_role	Topic	Original_excerpt
2	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Door design / requirements	Ja man designar efter kraven ... ständig förbättring och omdesign till dörren när kraven …
3	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Door design / requirements	Brandklasskrav är lagstadgade krav ... medan säkerhetsklass är en kommersiell klassning som gör det möjligt att sälja säkerhet eller en känsla av säkerhet till kunder.
4	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Requirements hierarchy	Och då är det två olika saker. Man kan bortse från säkerhetskraven om man uppfyller brandkraven, men man kan inte bortse från brandkraven för att erbjuda säkerhetskraven
5	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Material choice / fire performance	Material handlar framför allt om med vilket material man når den isoleringsförmåga som krävs, och vilken struktur som möter kraven på integritet vid brand
6	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Material choice / trade-offs	Ja, det måste vara med material som är tillgängliga, som finns i marknaden och som når en viss kommersiell kostnad som gör produkten intressant att sälja och köpa.
7	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Material choice / mineral wool	Mineralull väljs eftersom den är icke brännbar och har egenskaper som gör den lämplig ur brandsäkerhetssynpunkt.
8	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Material choice / mineral wool	Den är stabil, hjälper till med ljudisolering,är lätt att hantera, lätt att dimensionera och har ett rimligt pris
9	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Standardization	Det skiljer sig väldigt lite. Vi har siktat på att göra en produkt som når alla kraven som vi vill ha i vår dörr.
10	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Standardization	Vi har utvecklat samma skal på dörren med olika mineralull. Vi har samma typ av dörr, exakt samma skal och olika klassning, så rent visuellt ser man ingen skillnad mellan en 15-, 30- och 60-minuters brandklassad dörr.
11	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Standardization	Ja, det handlar om tjocklek på olika delar som isolering och förstärkningar och den typen av saker.

12	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Assembly method	Nitning ... Är ett enkelt fästningsmedel som kan göras halvmanuellt. Det är ett av de starkaste sätten att binda ihop två plåtbitar, och eftersom plåtarna redan är laminerade kan man inte svetsa. Man kan inte lägga till någon process som innebär värme
13	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Circular redesign constraints	Om du ändrar till exempel från nitning till svetsning eller limning, då måste testa om ... det är inte samma produkt
14	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Future design options	Om man ... bocka plåtarnas kanter ... så att man kan gifta ihop dem ... bara trycka plåten mot plåten
15	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Repair / reuse	Absolut och det gör vi idag ... lyfta dörrbladet, byta gångjärn och använda dörren igen
16	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Disassembly / recyclability	Vi har lyckats att använda ingen lim ... vi limmar inte
17	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Disassembly practicality	Det är rätt så enkelt ... fem sekunder
18	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	End-of-life limitation	Det skulle vara laminat. jag vet inte processerna för att kunna dela den laminatbehandling som man gör på ytan av plåtarna och kunna återanvända stål
19	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Material constraints	Det kan vara hur miljövänligt som möjligt och hur återanvändbart som möjligt. Men den når inte kraven vid tillverkningsprocess, så det måste vara en utveckling som levererar samma egenskaper som de som vi använder idag, eller att man kan nå dem i alla fall.
20	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	System perspective	Inte dörren i sig, men dörren är en produkt som är en del av en kedjeprocess, från inköp av råmaterial till transport.
21	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Packaging reuse	För att påverka den där bättre skulle vi kunna till exempel återanvända packningsmaterial, som idag gör vi väldigt lite av, i form av de pallar vi använder för att skicka dörrarna.
22	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	System perspective	Cirkularitet i processen borde innebära inte bara själva tillverkningen utan hela funktionskedjan.
23	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	System perspective	Cirkularitet ... inte bara själva tillverkare utan hela funktionskedja

24	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Actor alignment / policy	Det måste finnas överenskommelse mellan alla aktörer ... krav från kunden ... inställning av de som tillverkar ... lagligt krav
25	INT01	Produktions-/teknikansvarig hos ståldörrstillverkare	Redesign constraints	Vi kan inte ändra som vi vill ... om vi gör den där vi måste starta den där processen igen
26	INT02		Replacement drivers	De gamla dörrarna som sitter i fastighetsbeståndet i Sverige uppfyller inte dagens krav på gastäthet, brandklass och ljudisolering.
27	INT02		Replacement drivers	Man byter alltså till en modern ytterdörr för att uppfylla kraven på inbrottsskydd, brand, ljud och gastäthet.
28	INT02		Reuse practice	Det är väldigt få trädörrar som återbrukas, utan de tas tillbaka och sorteras på avfallsstationen.
29	INT02		Waste handling	Allt trä sorteras i träavfall ... metalldetaljer sorteras separat på avfallsstationen.
30	INT02		Reuse possibilities	Om det är en ståldörr ... den dörren kan man ju i vissa lägen återbruka, det vill säga montera någon annanstans.
31	INT02		Reuse possibilities	Då kan vi sälja dem eller skänka dem till räddningsverket ... som kan öva på att bryta upp sådana här dörrar.
32	INT02		Reuse limitations	Det är väldigt, väldigt sällan ... Än så länge är det inte så många ståldörrar som byts ut …
33	INT02		Replacement drivers	Nu vill vi ha enhetliga dörrar ... då blir det ju ett antal säkerhetsdörrar över.
34	INT02		Disassembly conditions	En ståldörr är ju enkel, för den är ju fastskruvad, så den kan man skruva bort hela karmen på.
35	INT02		Disassembly conditions	De gamla trädörrarna är ju spikade, och där måste vi såga sönder karmen …
36	INT02		Certification constraints	Dörrarna ... ska uppfylla de här kraven ... vara godkända av ett ackrediterat organ …
37	INT02		Reuse limitations	Om de inte uppfyller dagens krav så får de användas till något annat ... till exempel att öva på …

38	INT02	Repair limitations	Det är inte mycket poäng i att laga sådana här gamla dörrar ... Vi lagar inga gamla trädörrar …
39	INT02	Waste handling	Vi levererar dörrarna som de är till avfallsstationen.
40	INT02	Traceability	Ja, det antar jag, men jag vet faktiskt inte vad som händer.
41	INT03	Circularity work	Vi jobbar dels med Miljöbyggnadscertifieringar skulle jag säga i större byggprojekt. Om byggnaden då ska certifieras så finns det en indikator där som heter cirkularitet och då måste man liksom ta hand om den.
42	INT03	Reuse assessment	Förutom det som är i certifieringarna så kan man ju ta tag i återbruket ändå på lite olika sätt och göra återbruksinventering... Då gör vi det i CCBuild... fortfarande då i dialog med projektet som kanske redan har någon projektör inkopplad arkitekt.
43	INT03	Documentation / traceability	Det är ju viktigt med information så det blir kvalitet i det som finns fakta omkring och om det finns garantier kvar och vad det är för något historiken så att man också hittar om det kan finnas farliga ämnen.
44	INT03	Material constraints	Jag tar ju inte in några gamla ståldörrar. Det kan vara asbest i. Det är inte en chans att jag tar in där till mig... Årtal är jätteviktigt, annars måste man provta.
45	INT03	Repair practice	Gör rent och testar. Det är de som funkar. Riktar karmar ibland om de är lite skeva i stål. Målar upp dem... Byte av beslag och sådant... Det är ju inga stora ingrepp på en dörr jag håller på med.
46	INT03	Replacement drivers	Det är ju säkerhet och brand och det är mycket det som gör att vi ändå måste köpa nya.
47	INT03	Certification constraints	Om man då renoverar en lokal så kan man ju titta på om det finns något ställe där det inte behöver vara samma brandklass. Då kan vi återbruka den dörren. Om det inte är något farliga ämnen…
48	INT03	Certification constraints	Det måste ändå slutgodkännas. Till syvende och sist det är både bygglov och egenkontrollen där kontrollansvarig lagstiftning så kollar lagstiftningen. Så måste vi ju förhålla oss till det så att därför är det lite svårt med de där dörrarna.
49	INT03	Reuse / recycling pathways	Stål går väl som metall, då skickar vi den till metallåtervinning... och annars har ju vi också CCBuild, det som är OK, det kan vi lägga ut på en yttre marknadsplats... Och vi har också kontakt med ett företag som heter Klaravik som auktionerar ut.
50	INT03	Disassembly conditions	På byggen är de ju där vill de helst bara slänga allt i containern... Nej, det gör de ingen sådan, de har inte tid... I sådana fall måste du kravställa in upphandling.
51	INT03	Reuse barriers	Det kräver mycket kunskap... det kostar då pengar om vi måste undersöka det. Det kostar pengar med personal som kollar upp, tid. Tid och pengar och ämnen…
52	INT03	Documentation / traceability	Ja all den här informationen. Det är väl jättebra. Det sitter på en QR kod... får man direkt fakta... Ja, det kanske är användbart vid inventering och såna grejer också.
53	INT03	Circular design requirements	Det är väl att alltid kunna få informationen... ståldörrar som är lätta att ta isär, inga farliga ämnen, bra uppmärkning liksom... demonterbarhet, flexibilitet…
54	INT03	Joining method	Ja alltså så är det jätteenkelt... en vass borr i... metallborr... det går i alla fall.

Part 2:

Cleaned_meaning	Tentative_1st_order_concept	Memo	Merge Note	ROW
Dörren utvecklas utifrån externa krav och omdesignas när nya krav tillkommer	Vi designar dörren efter krav	He explains that door development is driven by external requirements and that the design is continuously revised when new requirements arise.	M1	2
Brandklasskrav är lagstadgade krav, medan säkerhetsklass är en kommersiell klassning som gör det möjligt att sälja säkerhet eller en känsla av säkerhet till kunder.	Brandkrav är lagstyrda, medan säkerhetskrav är kommersiellt drivna.	He clearly distinguishes between regulatory requirements and commercial requirements.	M1	3
Brandkrav och säkerhetskrav är två olika saker. Man kan bortse från säkerhetskraven om man uppfyller brandkraven, men man kan inte bortse från brandkraven för att erbjuda säkerhetskraven.	Brandkraven går före säkerhetskraven	He distinguishes between fire requirements and security requirements and shows that fire requirements take priority.	M1	4
Materialvalet styrs främst av vilka material och vilken konstruktion som kan uppfylla kraven på isolering och integritet vid brand.	Materialet väljs för att klara brandkraven.	Material choice is driven by fire requirements for insulation and integrity.	M1	5
Material måste både uppfylla prestandakrav och vara kommersiellt rimliga	Materialet måste både klara krav och pris	He explains that materials must be available on the market and affordable enough to make the product commercially viable.	M2	6
Mineralull används främst för att den uppfyller viktiga brandrelaterade krav.	Mineralull väljs för sina brandsäkra egenskaper.	He explains that mineral wool is chosen mainly because it supports fire safety.	M2	7
Mineralull väljs också eftersom den uppfyller praktiska och kommersiella krav.	Mineralull uppfyller även praktiska och kommersiella krav.	He points to stability, sound insulation, ease of handling, and cost as reasons for choosing mineral wool.	M2	8
Dörrtyperna skiljer sig lite eftersom företaget har strävat efter att utveckla en lösning som uppfyller flera krav inom samma grundprodukt.	De olika dörrtyperna skiljer sig väldigt lite.	He explains that the different door types are intentionally kept very similar.	M3	9
Företaget använder samma dörrskal för flera brandklasser, medan klassningen varieras genom andra interna lösningar.	De använder samma dörrskal för olika brandklassningar.	He explains that different fire ratings are built on the same basic door shell.	M3	10
Skillnaderna mellan brandklasserna ligger främst i isolering, förstärkningar och andra interna detaljer	Skillnaderna ligger i isolering och förstärkningar, inte i själva skalet.	He explains that the differences between door classes are mainly internal rather than visual or structural at the outer-shell level.	M3	11

Nitning används eftersom den är stark, enkel och inte skadar laminerade plåtar	Nitning väljs eftersom det är starkt och fungerar med laminerade plåtar.	He explains that riveting is chosen because it is a strong and practical joining method, and welding is not possible with laminated sheet metal.	M4	12
Byte av fogmetod kräver omtestning och ny klassificering	Ändrar vi fogmetod måste dörren testas om	Changing the joining method requires re-testing because it creates a different product.	M1	13
En möjlig framtida lösning är geometrier som låser ihop plåtar utan traditionell fogning	Det kan gå att låsa ihop plåtar genom formningen	He suggests a possible alternative joining method based on interlocking sheet-metal edges.	M4	14
Vissa delar kan redan idag repareras/bytas och dörren återanvändas	Vi reparerar redan idag genom att byta delar	He explains that some repair and reuse practices already exist in the current door system.	M5	15
Dörren innehåller i princip inga limmade förband i huvudkonstruktionen	Vi undviker lim i dörrkonstruktionen	He explains that glue is avoided in the door assembly.	M4	16
Nitar kan borras bort relativt enkelt och används även vid omarbete	Nitar går snabbt att borra bort	He explains that rivets can be removed relatively easily in practice.	M5	17
Laminatytan verkar vara ett hinder eftersom det är oklart hur ytbehandlingen kan separeras från stålet för återanvändning.	Laminatet försvårar återanvändning av stålet.	He explains that the laminate surface treatment may hinder steel reuse because it is difficult to separate from the steel.	M2	18
Ett nytt material räcker inte att det är miljövänligt och återanvändbart; det måste också fungera i tillverkningsprocessen och ge samma nödvändiga egenskaper som dagens material.	Ett mer hållbart material måste också fungera i tillverkningen	He explains that sustainability improvements must also meet manufacturing requirements.	M2	19
Dörren ses som en del av en större process, inte som en isolerad produkt.	Dörren är en del av en större kedjeprocess.	He explains that the door should be understood as part of a wider process rather than as a standalone product.	M6	20
Ett konkret sätt att öka cirkulariteten är att återanvända emballage och pallar i större utsträckning.	Vi kan öka cirkulariteten genom att återanvända emballage och pallar.	He points to packaging and pallet reuse as a concrete circularity improvement.	M6	21
Cirkularitet bör omfatta hela funktionskedjan, inte bara tillverkningen.	Cirkularitet handlar om hela funktionskedjan.	He takes a whole-chain view of circularity rather than limiting it to manufacturing.	M6	22
Cirkularitet ses som en fråga för hela kedjan, inte bara själva dörren	Cirkularitet handlar om hela kedjan	He explains that circularity should be understood at the whole-chain level, not only at the product level.	M6	23

Cirkularitet kräver samordning mellan kund, tillverkare och reglering	Alla aktörer måste dra åt samma håll	He explains that circularity depends on alignment between the actors in the process.	M6	24
De kan inte ändra produkten fritt, eftersom förändringar kräver att processen för provning och godkännande startas om.	Produktförändringar kräver att test- och godkännandeprocessen görs om.	He explains that design changes are constrained because they require the testing and approval process to start over.	M1	25
Gamla dörrar byts främst ut eftersom de inte uppfyller dagens krav på brand, ljud och gastäthet.	Gamla dörrar byts ut eftersom de inte uppfyller dagens krav.	He explains that door replacement is mainly driven by updated performance requirements.	M1	26
				27
Trädörrar återbrukas sällan och går oftast till avfallssortering.	Trädörrar återbrukas sällan.	He explains that removed wooden doors are rarely reused and usually become waste.	M6	28
Det demonterade materialet sorteras i olika avfallsfraktioner som trä, metall, deponi och övrigt avfall.	Rivningsmaterialet sorteras i separata avfallsfraktioner.	He explains that removed materials are sorted into separate waste streams.	M6	29
Ståldörrar kan i vissa fall återbrukas och användas i ett annat sammanhang.	Ståldörrar kan ibland återbrukas.	He explains that some steel doors can be reused instead of discarded.	M5	30
En praktisk form av återbruk är att använda gamla ståldörrar som övningsobjekt för räddningstjänst och polis.	Gamla ståldörrar kan återbrukas som övningsdörrar.	He points to emergency-service training as a concrete reuse pathway for old steel doors.	M5	31
Återbruk av ståldörrar är fortfarande ovanligt eftersom relativt få sådana dörrar ännu byts ut.	Återbruk av ståldörrar är fortfarande ovanligt	He explains that steel-door reuse is still rare because few such doors are currently being replaced.	M5	32
Dörrar kan också bytas ut för att skapa enhetlighet i fastigheten, även om de fortfarande fungerar.	Enhetlighet i fastigheten kan leda till att fungerande säkerhetsdörrar byts ut.	He explains that some functioning steel doors are replaced for reasons of visual or organizational uniformity.	M6	33
Ståldörrar kan demonteras i sin helhet eftersom de är fastskruvade.	Ståldörrar kan demonteras i sin helhet.	He explains that steel doors are relatively easy to remove intact because they are screwed in place.	M5	34
Trädörrar måste ofta sågas sönder vid demontering och kan därför inte tas ut hela.	Trädörrar förstörs ofta vid demontering.	He explains that wooden doors are often damaged during removal because the frame has to be cut apart.	M5	35
Nya dörrar måste vara certifierade enligt aktuella krav, vilket begränsar vilka dörrar som kan återanvändas i bostäder.	Certifieringskrav begränsar återanvändning av dörrar i bostäder.	He explains that certification requirements limit the reuse of old doors in residential settings.	M1	36
Äldre ståldörrar som inte uppfyller dagens krav bör återbrukas i andra användningar än bostäder.	Dörrar som inte uppfyller dagens krav får användas till andra ändamål.	He explains that non-compliant doors may still be reused, but only in alternative applications.	M1/M5	37

Gamla trädörrar repareras inte, eftersom det inte anses meningsfullt i förhållande till att byta till moderna dörrar.	Gamla trädörrar repareras inte utan byts ut.	He explains that old wooden doors are not repaired but replaced.	M5	38
De demonterade dörrarna separeras inte ytterligare av installatören utan skickas vidare som hela enheter till	Vi skickar dörrarna vidare som de är till avfallsstationen.	He explains that the company does not separate door materials further before sending them to the waste station.	M6	39
Det finns begränsad insyn i vad som händer med materialen efter att de har lämnats till avfallsstationen.	Vi vet inte vad som händer med materialen efter avfallsstationen.	He points to limited traceability after the doors leave the company's handling process.	M6	40
Cirkularitet hanteras delvis genom miljöbyggnadscertifieringar i större projekt.	Cirkularitet drivs delvis genom certifieringskrav.	She explains that circularity is partly addressed through building certification processes.	M6	41
Återbruksinventeringar görs i dialog med projektörer, beställare och projektledare, bland annat i CCBuild.	Återbruk bedöms genom inventering i dialog med projektet.	They describe reuse assessment as a collaborative process involving inventories and project dialogue.	M6	42
Information om historik, garantier och möjliga farliga ämnen är viktig för att kunna bedöma om något kan återbrukas.	Bra information behövs för att kunna bedöma återbruk.	They stress that documentation and historical information are crucial for reuse decisions.	M6	43
Gamla ståldörrar tas inte gärna in för återbruk eftersom det kan finnas asbest och då krävs årtalsbedömning eller provtagning.	Gamla ståldörrar är svåra att återbruka på grund av risk för asbest.	They explain that old steel doors are difficult to reuse because of possible asbestos contamination.	M2	44
Underhåll består främst av rengöring, testning, riktning av karmar, målning och byte av beslag eller handtag.	De gör främst mindre underhållsåtgärder på dörrar.	He explains that their work mainly involves small maintenance actions rather than major door rebuilding.	M5	45
Dörrar byts ofta ut eftersom nya krav på brand och säkerhet gör att man måste köpa nya.	Nya brand- och säkerhetskrav driver dörrbyten.	They explain that updated fire and safety requirements often force replacement rather than repair.	M1	46
En dörr kan återbrukas bara om den passar i ett sammanhang där rätt brandklass inte krävs, eller om den fortfarande uppfyller kraven och inte innehåller farliga ämnen.	Återbruk kräver att dörren passar rätt klassning och inte innehåller farliga ämnen.	They explain that reuse depends on classification fit and the absence of hazardous substances.	M1	47
Även om man vill återbruka en dörr måste lösningen slutgodkännas och följa bygglov, egenkontroll och annan lagstiftning.	Återbruk begränsas av slutgodkännande och lagkrav.	She explains that reuse is constrained by formal approval processes and regulatory compliance.	M1	48
Ståldörrar går främst till metallåtervinning, men dörrar som är i tillräckligt bra skick kan eventuellt läggas ut för återbruk via CCBuild. Klaravik nämns snarare som en kanal för andra objekt än specifikt dörrar.	Metallåtervinning är huvudspåret, medan CCBuild kan vara en möjlig återbruksväg.	They describe metal recycling as the main route, while CCBuild appears to be a possible reuse option. Klaravik is mentioned more as a platform for other items than specifically for doors.	M5	49
På byggplatser finns sällan tid att plocka isär dörrar mer noggrant; om det ska göras måste det kravställas i upphandling.	Demontering i detalj sker bara om det kravställs särskilt.	They explain that detailed disassembly usually does not happen unless it is explicitly required in procurement.	M6	50
Återbruk av ståldörrar hindras av behov av kunskap, undersökningar, personalresurser, tid och kostnader.	Återbruk hindras av tid, pengar och kunskap.	They identify time, cost, and knowledge requirements as major barriers to reuse.	M6	51
QR-koder eller annan enkel märkning med fakta om dörren skulle underlätta inventering, klassning och återbruk.	Enkel tillgång till dörrinformation skulle underlätta återbruk.	They see easily accessible product information as highly valuable for reuse and inventory work.	M6	52
De önskar dörrar som är demonterbara, flexibla, tydligt uppmärkta och fria från farliga ämnen, eller där farliga delar är lätta att ta bort och ersätta.	Cirkulära dörrar bör vara demonterbara, väldokumenterade och fria från farliga ämnen.	They describe better information, safe materials, and easy disassembly as key features of more circular doors.	M2	53
Demontering av popnitar beskrivs som möjlig och relativt enkel med en metallborr.	Popnitar kan tas bort relativt enkelt med metallborr.	He explains that pop rivets can be removed relatively easily using a metal drill bit.	M4	54

Appendix B2: Merge Groups and Theme Development

This table shows how the tentative first-order concepts were sorted into merge groups. It also explains why the codes were merged, identifies possible second-order themes, and links each group to the relevant research question.

Due to the width of the table, Appendix B2 is also presented in two parts. The two parts should be read together as one continuous table. The row number is included in both parts for reference.

Part 1:

ROW	Merge	Tentative_1st_order_concept	Merged 1st-order concepts	Why merged
2	M1	Vi designar dörren efter krav	Requirements shape the door design	This shows that the door design is driven by external requirements rather than free design choices.
3	M1	Materialet väljs för att klara brandkraven	Requirements shape the door design	This also shows that requirements directly shape design decisions, here through material choice.
4	M1	Nya brand- och säkerhetskrav driver dörrbyten	Requirements shape the door design	This shows that updated requirements affect replacement decisions and the continued use of door types.
5	M1	Brandkrav är lagstyrda, medan säkerhetskrav är kommersiellt drivna	Fire requirements have a distinct regulatory status	This distinguishes between legally required and commercially driven requirements.
6	M1	Brandkraven går före säkerhetskraven	Fire requirements have a distinct regulatory status	This shows that fire requirements take priority over other requirement types.
7	M1	Ändrar vi fogmetod måste dörren testas om	Certification and approval constrain design changes	This shows that even a change in joining method triggers re-testing and limits redesign freedom.
8	M1	Produktförändringar kräver att test- och godkännandeprocessen görs om	Certification and approval constrain design changes	This expresses the same logic more generally: product changes require a new approval process.
9	M1	Certifieringskrav begränsar återanvändning av dörrar i bostäder	Certification and approval constrain reuse	This shows that compliance requirements restrict where older doors can be reused.
10	M1	Återbruk begränsas av slutgodkännande och lagkrav	Certification and approval constrain reuse	This shows that reuse depends on formal approval and regulatory requirements.
11	M1	Återbruk kräver att dörren passar rätt klassning och inte innehåller farliga ämnen	Reuse requires the right classification fit and compliance fit	This shows that reuse is only possible when the door fits the required classification context and can be accepted in that setting.
12	M2	Materialen måste vara marknadstillgängliga och kommersiellt rimliga	Material selection must be commercially viable	This shows that materials must be available on the market and affordable enough to make the product viable to sell and buy.
13	M2	Det svåra är att kombinera kravuppfyllelse med säljbarhet	Material selection must be commercially viable	This expresses the same logic more broadly: meeting requirements is not enough unless the product is also commercially viable.
14	M2	Mineralull väljs för sina brandsäkra egenskaper	Mineral wool is chosen because it satisfies multiple requirements	This shows that mineral wool is selected because it supports fire safety and required technical performance.
15	M2	Mineralull uppfyller även praktiska och kommersiella krav	Mineral wool is chosen because it satisfies multiple requirements	This adds that mineral wool is not only technically suitable, but also practical and commercially acceptable.
16	M2	Ett mer hållbart material måste också fungera i tillverkningen	More sustainable materials must also work in manufacturing and performance terms	This shows that more sustainable alternatives are only relevant if they can be processed in production and still deliver the required properties.

17	M2	Laminatet försvårar återanvändning av stålet	Surface treatments and material combinations hinder reuse and recycling	This shows that the laminate surface treatment makes it difficult to separate and reuse the steel at end of life.
18	M2	Gamla ståldörrar är svåra att återbruka på grund av risk för asbest	Hazardous substances hinder reuse and material handling	This shows that suspected hazardous substances, such as asbestos, make reuse more difficult and create additional handling requirements.
19	M2	Cirkulära dörrar bör vara demonterbara, väldokumenterade och fria från farliga ämnen	Future material choices should support safe and circular handling	This points toward future material-related design recommendations: safe materials, clear information, and easier end-of-life handling.
20	M3	De olika dörrtyperna skiljer sig väldigt lite	Different door classes are built on a shared basic structure	This shows that CH Industry's different door classes are intentionally kept very similar rather than being redesigned from scratch for each class.
21	M3	De använder samma dörrskal för olika brandklassningar	Different door classes are built on a shared basic structure	This shows more specifically that CH Industry uses the same door shell across several fire classifications.
22	M3	Skillnaderna ligger i isolering och förstärkningar, inte i själva skalet	Variation between door classes lies mainly in internal components	This shows that differences between CH Industry's door classes are created mainly through internal components, such as insulation and reinforcements, rather than through the outer shell.
23	M4	Nitning väljs eftersom det är starkt och fungerar med laminerade plåtar	Riveting is used because it works well with the current door construction	This shows that riveting is chosen because it is strong, practical, and compatible with laminated sheet metal that cannot easily be welded.
24	M4	Vi undviker lim i dörrkonstruktionen	Current joining relies on mechanical methods rather than adhesive bonding	This shows that the current door construction avoids glue-based bonding and instead relies on mechanical assembly.
25	M4	Det kan gå att låsa ihop plåtar genom formningen	Alternative joining solutions may be possible in future door design	This points to a possible future design solution where sheet-metal parts could be joined through form or geometry rather than through a separate fastening method.
26	M4	Popnitar kan tas bort relativt enkelt med metallborr	Pop riveting is not seen as a major barrier to disassembly	This shows that pop rivets are considered relatively easy to remove in practice and therefore not a major obstacle to disassembly.
27	M5	Vi reparerar redan idag genom att byta delar	Some repair and reuse already take place in the current door system	This shows that certain components can already be replaced and the door can continue to be used rather than discarded immediately.
28	M5	De gör främst mindre underhållsåtgärder på dörrar	Some repair and reuse already take place in the current door system	This shows that current practice includes minor maintenance and functional upkeep rather than only replacement.
29	M5	Ståldörrar kan ibland återbrukas	Steel doors have some reuse potential	This shows that removed steel doors can sometimes be reused instead of directly becoming waste.

30	M5	Gamla ståldörrar kan återbrukas som övningsdörrar	Steel doors have some reuse potential	This gives a concrete example of how old steel doors can be reused in practice, even if not in their original residential function.
31	M5	Återbruk av ståldörrar är fortfarande ovanligt	Reuse of steel doors is still limited in practice	This shows that although reuse is possible, it is still uncommon in current practice.
32	M5	Ståldörrar kan demonteras i sin helhet	Steel doors are practically easier to disassemble and handle than wooden doors	This shows that screwed steel doors can be removed intact, which supports reuse and handling.
33	M5	Trädörrar förstörs ofta vid demontering	Steel doors are practically easier to disassemble and handle than wooden doors	This provides a contrast showing that wooden doors are often damaged during removal, unlike steel doors that can be removed more intact.
34	M5	Gamla trädörrar repareras inte utan byts ut	In practice, older doors are often replaced rather than repaired	This shows that repair is often not seen as meaningful for older doors, especially old wooden doors.
35	M5	Ståldörrar går till återbruk om de är OK, annars till metallåtervinning	The condition of the door determines whether it is reused or recycled	This shows that practical handling depends on whether the removed door is good enough for reuse; otherwise it becomes material recycling.
36	M6	Dörren är en del av en större kedjeprocess	Circularity must be understood in a broader system context	This shows that the door is not treated as an isolated product, but as part of a larger chain including sourcing, transport, installation, and end-of-life handling.
37	M6	Cirkularitet handlar om hela funktionskedjan	Circularity must be understood in a broader system context	This explicitly frames circularity as something that concerns the full functional chain rather than only the manufacturer or the product.
38	M6	Cirkularitet handlar om hela kedjan	Circularity must be understood in a broader system context	This expresses the same broader systems view of circularity across the product chain.
39	M6	Vi kan öka cirkulariteten genom att återanvända emballage och pallar	Circularity is also shaped by logistics and surrounding flows	This shows that circular improvement is not limited to the door itself, but also includes packaging, pallets, and transport-related handling.
40	M6	Alla aktörer måste dra åt samma håll	Circularity requires coordination between multiple actors	This shows that circularity depends on alignment between customers, manufacturers, and other actors in the process.
41	M6	Kunden måste ställa krav på cirkularitet	Circularity requires coordination between multiple actors	This shows that customer demand is one of the drivers needed for circular solutions to become relevant in practice.
42	M6	Det behövs lagkrav och politisk press för att driva cirkularitet	Circularity requires coordination between multiple actors	This shows that policy and regulation are also necessary drivers, alongside market actors, to make circularity happen in practice.
43	M6	Bra information behövs för att kunna bedöma återbruk	Accessible information is crucial for reuse	This shows that reuse decisions depend on having reliable information about the product, its history, and its condition.
44	M6	Enkel tillgång till dörrinformation skulle underlätta återbruk	Accessible information is crucial for reuse	This shows that easier access to product information, such as via QR codes or product passports, could improve reuse and inventory work.
45	M6	Vi vet inte vad som händer med materialen efter avfallsstationen	Limited visibility makes follow-up difficult after waste handling	This shows that traceability is weak once the doors leave the company's own handling process.
46	M6	Demontering i detalj sker bara om det kravställs särskilt	Practical circularity depends on procurement, time, and resource planning	This shows that more detailed disassembly does not happen by default, but must be specified and planned for in procurement and project execution.
47	M6	Återbruk hindras av tid, pengar och kunskap	Practical circularity depends on procurement, time, and resource planning	This shows that reuse is constrained not only by technical issues, but also by competence, cost, labor, and time.
48	M6	Återbruk bedöms genom inventering i dialog med projektet	Practical circularity depends on procurement, time, and resource planning	This shows that reuse work is embedded in project processes, coordination, and early planning rather than handled separately at the end.

Part 2:

Possible 2nd-order theme	Related RQ	ROW
Regulatory and performance requirements shape current door design	RQ1	2
Regulatory and performance requirements shape current door design	RQ1	3
Regulatory and performance requirements shape current door design	RQ1	4
Regulatory and performance requirements shape current door design	RQ1	5
Regulatory and performance requirements shape current door design	RQ1	6
Compliance requirements constrain redesign and reuse	RQ1	7
Compliance requirements constrain redesign and reuse	RQ1	8
Compliance requirements constrain redesign and reuse	RQ1	9
Compliance requirements constrain redesign and reuse	RQ1	10
Compliance requirements constrain redesign and reuse	RQ1/RQ2	11
Material selection is shaped by performance, practicality, and commercial viability	RQ1	12
Material selection is shaped by performance, practicality, and commercial viability	RQ1	13
Material selection is shaped by performance, practicality, and commercial viability	RQ1	14
Material selection is shaped by performance, practicality, and commercial viability	RQ1	15
Material selection is shaped by performance, practicality, and commercial viability	RQ1/RQ2	16
Material composition constrains end-of-life circularity	RQ1/RQ2	17
Material composition constrains end-of-life circularity	RQ1	18
Future material choices should enable safer and more circular door systems	RQ2	19
Standardized door architecture supports multiple classifications	RQ1	20
Standardized door architecture supports multiple classifications	RQ1	21
Standardized door architecture supports multiple classifications	RQ1	22
Current joining methods are shaped by production and material conditions	RQ1	23
Current joining methods are shaped by production and material conditions	RQ1	24
Future joining solutions could support more circular door design	RQ2	25
Current joining methods do not necessarily prevent disassembly	RQ1/RQ2	26
Existing practices already enable limited circularity in use and maintenance	RQ1	27
Existing practices already enable limited circularity in use and maintenance	RQ1	28
Steel doors offer limited but real reuse opportunities	RQ1	29

Steel doors offer limited but real reuse opportunities	RQ1/RQ2	30
Steel doors offer limited but real reuse opportunities	RQ1	31
Disassembly conditions influence practical reuse potential	RQ1/RQ2	32
Disassembly conditions influence practical reuse potential	RQ1	33
Current maintenance practice favors replacement over repair in older door systems	RQ1	34
Practical reuse depends on condition and handling possibilities	RQ1/RQ2	35
Circularity depends on system-level conditions beyond the door itself	RQ1	36
Circularity depends on system-level conditions beyond the door itself	RQ1	37
Circularity depends on system-level conditions beyond the door itself	RQ1	38
Circularity depends on system-level conditions beyond the door itself	RQ1/RQ2	39
Circularity depends on actor alignment and coordination	RQ1/RQ2	40
Circularity depends on actor alignment and coordination	RQ1/RQ2	41
Circularity depends on actor alignment and coordination	RQ1/RQ2	42
Information availability shapes reuse decisions	RQ1	43
Information availability shapes reuse decisions	RQ1/RQ2	44
Information availability shapes reuse decisions	RQ1	45
Practical implementation depends on project conditions and resource allocation	RQ1/RQ2	46
Practical implementation depends on project conditions and resource allocation	RQ1/RQ2	47
Practical implementation depends on project conditions and resource allocation	RQ1	48

Appendix B3: Development of Second-Order Themes and Aggregate Dimensions

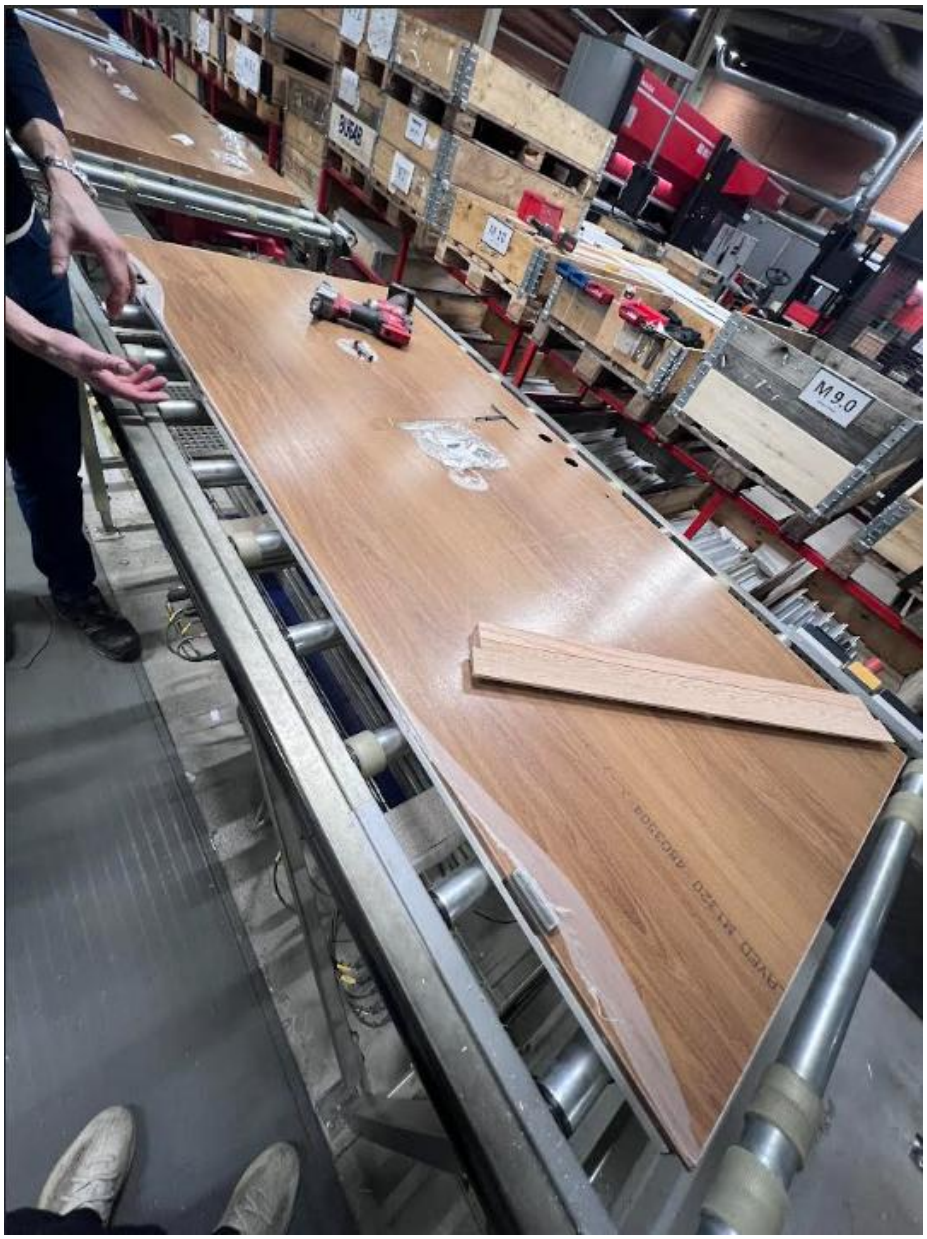
This table shows how the merged first-order concepts were developed into second-order themes and possible aggregate dimensions. It includes the rationale for each grouping and links each theme to the relevant research question.

This table formed the basis for the Gioia-inspired data structure figure presented in the results.

Merged 1st-order concept	2nd order theme	Rationale	Related RQ	Possible aggregate dimention
Requirements shape the door design	Current door design is governed by regulatory and performance requirements	This concept captures how current door design is shaped by external requirements rather than free design choices.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Fire requirements have a distinct regulatory status	Current door design is governed by regulatory and performance requirements	This concept captures the special priority and legal status of fire requirements in current door design.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Certification and approval constrain design changes	Compliance requirements constrain redesign and reuse	This concept shows that even relatively small design changes can trigger re-testing and approval demands.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Certification and approval constrain reuse	Compliance requirements constrain redesign and reuse	This concept shows that compliance requirements also limit where and how removed doors can be reused.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Reuse requires the right classification fit and compliance fit	Compliance requirements constrain redesign and reuse	This concept shows that reuse is not only about technical condition, but also about whether the door fits the required classification context.	RQ1/RQ2	Current steel door design is shaped by external requirements and industrial production logic
Material selection must be commercially viable	Material selection balances performance, practicality, and commercial viability	This concept captures that material choices are shaped not only by technical performance, but also by market availability and cost.	RQ1	Material choices both enable performance and constrain circularity
Mineral wool is chosen because it satisfies multiple requirements	Material selection balances performance, practicality, and commercial viability	This concept shows how one material is selected because it meets several technical, practical, and commercial needs at once.	RQ1	Material choices both enable performance and constrain circularity
More sustainable materials must also work in manufacturing and performance terms	Material selection balances performance, practicality, and commercial viability	This concept shows that future sustainable materials must still function in production and deliver required performance.	RQ1/RQ2	Material choices both enable performance and constrain circularity
Surface treatments and material combinations hinder reuse and recycling	Material composition constrains end-of-life circularity	This concept captures how current material combinations complicate separation, reuse, and recycling at end of life.	RQ1/RQ2	Material choices both enable performance and constrain circularity
Hazardous substances hinder reuse and material handling	Material composition constrains end-of-life circularity	This concept shows that hazardous substances create major barriers to reuse, handling, and safe recovery.	RQ1	Material choices both enable performance and constrain circularity

Future material choices should support safe and circular handling	Future material choices should enable safer and more circular door systems	This concept is clearly forward-looking and translates current material problems into future design implications.	RQ2	Future circular door systems require design for safer handling, disassembly, and information access
Different door classes are built on a shared basic structure	Standardized door architecture supports multiple classifications	This concept shows that multiple door classes are based on the same general construction logic.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Variation between door classes lies mainly in internal components	Standardized door architecture supports multiple classifications	This concept shows that differences between door classes are mainly created internally rather than through a completely different outer design.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Riveting is used because it works well with the current door construction	Current joining methods are shaped by production and material conditions	This concept captures how joining choices are adapted to the existing material setup and manufacturing logic.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Current joining relies on mechanical methods rather than adhesive bonding	Current joining methods are shaped by production and material conditions	This concept reflects the logic of the current assembly method and its practical implications.	RQ1	Current steel door design is shaped by external requirements and industrial production logic
Pop riveting is not seen as a major barrier to disassembly	Current joining methods do not necessarily prevent disassembly	This concept shows that the existing joining method may be more compatible with disassembly than initially assumed.	RQ1/RQ2	Existing practices and technical conditions create both circular opportunities and limitations
Alternative joining solutions may be possible in future door design	Future joining solutions could support more circular door design	This concept is clearly future-oriented and points toward design recommendations for improved circularity.	RQ2	Future circular door systems require design for safer handling, disassembly, and information access
Some repair and reuse already take place in the current door system	Existing practices already enable limited circularity in use and maintenance	This concept shows that current practices already contain some circular elements, even if limited.	RQ1	Existing practices and technical conditions create both circular opportunities and limitations
Steel doors have some reuse potential	Steel doors offer limited but real reuse opportunities	This concept captures that reuse is possible, but only under certain circumstances.	RQ1/RQ2	Existing practices and technical conditions create both circular opportunities and limitations
Reuse of steel doors is still limited in practice	Steel doors offer limited but real reuse opportunities	This concept adds that current reuse is still rare, even when technically possible.	RQ1	Existing practices and technical conditions create both circular opportunities and limitations
Steel doors are practically easier to disassemble and handle than wooden doors	Disassembly conditions influence practical reuse potential	This concept shows that disassembly conditions affect whether reuse is practically possible.	RQ1/RQ2	Existing practices and technical conditions create both circular opportunities and limitations
In practice, older doors are often replaced rather than repaired	Current maintenance practice favors replacement over repair in older door systems	This concept captures that the dominant practice is still replacement, not repair.	RQ1	Existing practices and technical conditions create both circular opportunities and limitations
The condition of the door determines whether it is reused or recycled	Steel doors offer limited but real reuse opportunities	This concept shows that practical pathways depend heavily on the condition of the removed door.	RQ1/RQ2	Existing practices and technical conditions create both circular opportunities and limitations
Circularity must be understood in a broader system context	Circularity depends on system-level conditions beyond the door itself	This concept frames circularity as a system issue rather than only a product issue	RQ1	Circularity is conditioned by broader organizational and system-level factors
Circularity is also shaped by logistics and surrounding flows	Circularity depends on system-level conditions beyond the door itself	This concept shows that packaging, transport, and surrounding flows also matter for circularity.	RQ1/RQ2	Circularity is conditioned by broader organizational and system-level factors
Circularity requires coordination between multiple actors	Circularity depends on actor alignment and coordination	This concept shows that circularity depends on coordination among customers, manufacturers, municipalities, and other actors.	RQ1/RQ2	Circularity is conditioned by broader organizational and system-level factors
Accessible information is crucial for reuse	Information availability shapes reuse decisions	This concept shows that reuse depends on knowing what the product is, what it contains, and whether it fits a new use context.	RQ1/RQ2	Circularity is conditioned by broader organizational and system-level factors
Limited visibility makes follow-up difficult after waste handling	Information availability shapes reuse decisions	This concept shows that traceability becomes weaker once the product leaves the actors directly involved in the project.	RQ1	Circularity is conditioned by broader organizational and system-level factors
Practical circularity depends on procurement, time, and resource planning	Practical implementation depends on project conditions and resource allocation	This concept shows that even technically possible circular practices depend on project conditions, procurement, time, and staffing.	RQ1/RQ2	Circularity is conditioned by broader organizational and system-level factors

Appendix C - Photos from production walkthrough.




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CERTIFIERING
KONTROLL
1002
RI.SE
Typ CHL 133
SERIE NR 202601292375
Brandklass EI2 15/EW 30
Brandgastäthetsklass S200
Ljudreduktion Rw 40 dB
Typg. Nr. SC1388-17
Säkerhetsdörr SS-EN 1627 RC3 SBSC Certifikat 18-122
SBSC

